Don't Oversample the Boundary: A Remedy Decomposition of the Minority Deficit in Imbalanced Learning

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Abstract

Class imbalance is overwhelmingly handled by oversampling, yet SMOTE-family methods typically trade majority-class accuracy for minority-class recall, and it is rarely clear *why* oversampling helps. We introduce a *remedy decomposition* of the misclassified minority: an out-of-bag error triage (ensemble confidence and local class representation, validated against an aleatoric–epistemic uncertainty decomposition) is combined with a threshold-recoverability test, assigning each minority error to the remedy that would fix it: *threshold-recoverable* (its score already clears the tuned decision threshold; only the default cutoff is wrong), *data-reducible* (more same-class data would help), or *irreducible* (genuine class overlap). Across 79 real datasets the deficit is predominantly threshold-recoverable (mean 68%; 80% at imbalance ratio > 3); only $\sim$6% is data-reducible and $\sim$17% irreducible. This explains a known but unexplained regularity: SMOTE-family resampling does not improve ranking (AUC is unchanged) and a threshold move reproduces its benefit without synthetic generation; single undertrained neural networks, whose ranking imbalance itself corrupts, are the exception. On controlled mixtures with a *known* Bayes boundary, standard SMOTE places an overlap-dependent fraction of synthetic points *across* that boundary, and a short proposition shows why interpolating boundary seeds crosses into majority-posterior space. Once every strategy receives the same out-of-fold threshold tuning, the default-threshold gap between non-generative and generative families collapses to within ± 1 pp across three base learners (against

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+6.4 pp from tuning the threshold alone): the SMOTE family's apparent edge is an operating-point effect, not a discrimination gain. The practical conclusion: diagnose the deficit; for most of it, move the decision rather than oversample.

Keywords: class imbalance, SMOTE, decision threshold, calibration, uncertainty decomposition, data complexity

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1. Introduction

Class imbalance is a persistent challenge in machine learning. In production settings the cost of the resulting performance degradation is not abstract: in medical screening at 10% prevalence, a one-percentage-point drop in specificity at fixed sensitivity multiplies the false-positive volume that clinicians must reconcile, and in fraud detection it directly inflates the analyst caseload that determines operational feasibility. When one class dominates the training distribution, classifiers tend to optimize for majority-class accuracy at the expense of minority-class recall [1]. SMOTE [2], which generates synthetic minority instances by interpolating between existing samples and their nearest neighbours, remains the dominant solution after two decades and has spawned hundreds of variants [3, 4]. Yet a stubborn regularity persists: SMOTE variants almost invariably degrade majority-class accuracy while improving minority-class recall. This accuracy–balanced-accuracy tradeoff is so common that practitioners treat it as inherent to oversampling.

The field has responded in two ways. One line asks *which* resampling method a given dataset needs: data-complexity studies show that data characteristics, not the imbalance ratio alone, moderate oversampling effectiveness [5], and meta-learning research recommends a resampler from dataset meta-features [6, 7]. Another line has grown openly critical: large benchmark studies find that oversampling often fails to outperform doing nothing once evaluation is done carefully, and that reported gains depend heavily on which metrics are chosen [8, 9]. Both lines are *correlational*: they relate measured data characteristics or benchmark outcomes to observed performance, but do not explain *why* a method helps or harms, and the proposed selection rules barely beat the trivial "always SMOTE" baseline. What is missing is a *mechanistic* account of the tradeoff that tells us which lever to pull.

We provide such a mechanistic account. We argue the tradeoff is *not* inherent to oversampling but is a mechanistic consequence of *where* SMOTE generates synthetic data. Standard SMOTE uses every minority instance as an interpolation seed, including those in regions of genuine class overlap near the decision boundary. Generating there pushes the learned boundary into majority-class territory, recovering some minority instances while misclassifying previously correct majority ones. We establish this in two steps. On controlled Gaussian mixtures with a *known* Bayes boundary, standard SMOTE demonstrably places synthetic points *across* that boundary. On real data, where the boundary is unobserved, a controlled interventional experiment manipulates *where* synthetic points are generated: augmenting *only* the model-estimated boundary (irreducible) errors reproduces the full SMOTE signature (balanced accuracy $p < 0.001$, negative accuracy trend), whereas augmenting non-boundary or random instances has no such effect. The evidence indicates that much of the cost traces to *where* synthetic data is placed, rather than to rebalancing per se.

To localize this boundary/overlap region on real data we introduce **Error Instance Triage**, which categorizes each misclassified training instance from *out-of-bag* random-forest confidence and local class representation; the categories are interpreted through, and validated against, an aleatoric–epistemic uncertainty decomposition [10, 11]. We state the scope plainly: the triage *instrument* is computed from random-forest ensembles (the standard vehicle for this decomposition [10]), while the downstream conclusions are evaluated across three base learners, gradient-boosted trees, random forests, and logistic regression (§5.5). The categories are:

1. **Cat 1 (Noise):** likely mislabeled instances: high total but low aleatoric uncertainty; the label conflicts with the feature pattern.
2. **Cat 2 (Data-limited):** errors in regions lacking same-class evidence: high epistemic uncertainty, low local class density. These are *reducible*: more correct-class data would resolve them.
3. **Cat 3 (Irreducible):** errors in estimated boundary/overlap regions where classes overlap: high aleatoric uncertainty. These are *not* reducible by adding data.

The decomposition yields a diagnostic, not a new oversampler. On real imbalanced data the misclassified minority concentrates in these estimated boundary/overlap regions (Cat 3 is $\sim$85% of errors, a fraction that is stable under strict out-of-fold re-estimation), exactly where SMOTE's harm originates. Yet that balanced-accuracy deficit turns out to be largely a *decision-threshold artefact*: moving the decision threshold alone (a non-generative change) recovers roughly nine percentage points of balanced accuracy over the default threshold (§5.5), without any synthetic generation. The practical implication is that the lever is the *decision rule*, not synthetic generation, and indeed non-generative strategies (threshold-moving, cost-sensitive learning, balanced ensembles) capture the recoverable balanced accuracy *without* the boundary-generation accuracy cost.

We make this precise with a *remedy decomposition* of the misclassified minority that sorts each error into threshold-recoverable, data-reducible (epistemic), or irreducible (aleatoric). We find the deficit is overwhelmingly **threshold-recoverable** (mean 68% across 79 datasets, 80% at IR > 3): the classifier's scores already place these instances above a tuned operating point. The known "move the threshold" equivalence [12, 13, 9, 14] is the consequence: across a 16-strategy menu oversampling yields no ranking gain and shifts the class prior in probability space rather than improving calibration, and once *every* strategy is given the same out-of-fold threshold tuning the large default-threshold gap between the families collapses to within ± 1 pp across three base learners. We turn the diagnosis into a prescription with a regime map and an honestly-scoped selector, which beats the always-SMOTE / imbalance-ratio paradigm, but only because the menu now includes the dominant non-generative strategies, while a strong non-generative default is itself near-optimal.

Contributions.

1. **A remedy decomposition of the minority deficit (headline).** We decompose the *misclassified* minority by combining an error triage (out-of-bag confidence and local representation, validated against an aleatoric–epistemic uncertainty decomposition) with a threshold-recoverability test, mapping each error to the remedy that would fix it: *threshold-recoverable*, *data-reducible* (epistemic), or *irreducible* (aleatoric). Unlike prior instance taxonomies, which score *difficulty* [15, 16, 17, 18], ours conditions on the model being wrong and introduces the threshold-recoverable axis. Empirically the deficit is overwhelmingly threshold-recoverable (68% on average, 80% at IR > 3).
2. **A mechanism behind the "move the threshold" equivalence.** The decomposition *explains* a known regularity (that resampling does not improve ranking and a threshold shift reproduces its benefit [12, 13, 9, 14]) by showing the deficit is mostly a thresholding artefact, so oversampling is an implicit, inefficient enactment of a threshold

move. We give the geometric mechanism: on controlled mixtures with a *known* boundary, SMOTE generates across it, and a short proposition (§3.4) shows why interpolating boundary seeds crosses into the majority-posterior region.

3. **Evidence at scale (confirmation, not claim).** Across 79 datasets and a 16-strategy menu we reproduce, on a broad roster, that oversampling yields no discrimination gain (AUC unchanged) and enacts a prior shift in probability space rather than a calibration improvement, and that once *every* strategy receives the same out-of-fold threshold tuning, the +5.76 pp default-threshold gap between the families collapses to within ±1 pp across three base learners (six evaluated; the documented exception is the single MLP, whose *ranking* imbalance corrupts and resampling partially repairs, §5.5). The apparent dominance of non-generative methods at the default threshold is an operating-point effect. We frame these as confirmation of [9, 14, 19] at scale.
4. **An honestly-scoped prescription.** The decomposition says which remedy applies; we add a regime map and, as a secondary analysis, a held-out selector that beats always-SMOTE by 3.5 points, but an imbalance-ratio-only rule matches it and a strong non-generative default is near-optimal, so the *menu*, not selector sophistication, is the lever (the practical answer to the selection problem left open by [7]).

2. Related Work

Instance categorization for imbalanced learning. Several frameworks decompose training data by learning difficulty. Instance Hardness [16] assigns a difficulty score per instance from majority votes across multiple learners. Dataset Cartography [17] maps instances by training-dynamics confidence/variability. Data-IQ [18] extends this to tabular data via prediction trajectories. The closest line of work is Napierała & Stefanowski [20, 15], who categorize minority instances into safe, borderline, rare, and outlier based on the proportion of same-class neighbours in a k-nearest-neighbour graph; Sáez et al. [21] extend this to multi-class imbalanced data and analyze how oversampling interacts with each type. Crucially, all of these characterize instance *difficulty*; none condition on the model being *wrong* and ask *which remedy applies*. Our decomposition differs in three ways: (i) it operates on the *misclassified minority* and maps each error to a remedy (move the threshold, add data, or accept it) rather than scoring difficulty; (ii) it introduces a *threshold-recoverable* class (the model's score already places the instance above a tuned operating point; only the default cutoff is wrong), an axis absent from prior difficulty taxonomies; and (iii) it combines a model-based aleatoric–epistemic decomposition (separating Bayes-rate ambiguity from sample-size limits, which k-NN typing cannot) with a local class-ratio measure, used as a diagnostic rather than as a competing oversampler.

Threshold-moving, cost-sensitivity, and the limits of resampling. A long line of work argues the decision threshold, not the training distribution, is the right lever under imbalance. Provost [12] warns against the default 0.5 cutoff, and Elkan [13] shows that changing class proportions is *equivalent* to changing the decision threshold/costs, so rebalancing barely alters the learned scoring function; Drummond and Holte [22] reach the analogous conclusion via cost curves. Large-scale and clinical studies confirm the empirical consequence: resampling does not improve ranking/AUC and its apparent gains are reproduced by a threshold shift, while it degrades probability calibration [9, 14, 19, 23]. A parallel critical line reaches a compatible verdict from benchmarking: Tarawneh et al. [8] review the oversampling literature and argue that reported gains are often artefacts of evaluation choices, and Sağlam and Cengiz [24] observe that whether oversampling appears to help depends strongly on *which* metric is consulted. Our decomposition explains where this metric dependence comes from: an operating-point move registers on threshold-sensitive metrics (balanced accuracy, recall, G-mean) while leaving ranking metrics (AUC) unchanged, so the verdict on oversampling is inherited from the metric's threshold sensitivity. We do *not* claim these regularities as new; we reproduce them across our roster and instead *explain* them: our reducibility decomposition shows the misclassified minority is predominantly *threshold-recoverable*, which is *why* a threshold move suffices and oversampling behaves as an implicit, inefficient enactment of it. Our contribution is the per-instance instrument that accounts for this equivalence and locates its exceptions (the data-reducible and irreducible minority), not the equivalence itself.

Class overlap and imbalance. Santos et al. [25] provide a unifying multi-view panorama of the relationship between class overlap and class imbalance, arguing that overlap is often the underlying driver of imbalance-related difficulty. Their framing supports our boundary-generation diagnosis: when SMOTE generates near class-overlap regions, the harm to accuracy is a predictable consequence of overlap-blind seeding.

Characteristic-driven and complexity-guided selection. Komorniczak, Ksieniewicz, and Woźniak [5] demonstrate that data-complexity metrics correlate with the accuracy gain oversampling delivers, and Carvalho et al. [6] survey how resampler performance tracks data structure. Most directly, Jiang and Ye [7] control the imbalance ratio and show data characteristics moderate oversampler choice, yet their resulting selection rule does *not* beat always-SMOTE, and they leave open whether a held-out, characteristic-driven selector can. This line is *correlational*: it relates measured complexity to observed performance without explaining the mechanism. Our contribution is orthogonal (a mechanism-level account of *why* boundary generation hurts), and we answer the open selection question directly, on a full strategy menu with held-out validation (§5.7), finding that the menu, not the selector, is the dominant lever. Our account is mechanism-level and intervention-supported rather than correlational.

SMOTE variants and cleaning. Borderline-SMOTE [26] restricts generation to instances near the decision boundary; ADASYN [27] upweights generation in sparse minority regions. Both methods *increase* generation in exactly the regions we identify as harmful. Safe-Level-SMOTE [28] uses each seed's "safe level" (count of same-class neighbours) to gate interpolation, an idea closely related to ours but instantiated via k-NN heuristics rather than uncertainty decomposition; we compare directly in §5. Post-hoc cleaning methods combine SMOTE with instance removal: SMOTE+ENN [29, 30] removes instances misclassified by edited nearest neighbours, while SMOTE+Tomek [29, 31] removes Tomek-link pairs. These methods are agnostic to *why* an instance is problematic, removing both correctable and boundary instances indiscriminately. Beyond the canonical family, MWMOTE [32], ProWSyn [33], and polynomial-fitting SMOTE [34] represent more recent extensions. Geometric SMOTE [35] replaces linear interpolation altogether: it draws synthetic points inside a (possibly truncated and deformed) hypersphere around each minority seed whose radius can be bounded by the distance to the *nearest majority* instance, an explicit geometric guard against generating in the other class's territory, and hence a design-level response to precisely the boundary-crossing mass that Proposition 1(iii) identifies as the harmful quantity; we test it directly, on the controlled boundary and at scale (§5.3, §5.5). The 85-variant survey of Kovács [4] provides a broader perspective; we include the top three of his ranked variants as baselines. Theoretically, Sakho et al. [36] show that *default* SMOTE asymptotically copies minority points and that its synthetic density *vanishes* near the boundary of the minority support; our boundary-crossing account is therefore specific to the *overlap* regime: interpolation between minority seeds that already lie in class-overlap territory, where the segment crosses into majority-posterior space (Proposition 1). It is complementary to, not in conflict with, their clean-margin analysis.

Uncertainty-guided oversampling. A distinct line replaces geometric seeding heuristics with *uncertainty* signals that decide where to generate. Ertekin [37] adaptively oversamples the informative region near the decision boundary; CGMOS [38] weights candidate seeds by the certainty change a synthetic point would induce on both classes; Aguiar and Cano [39] allocate an active-learning budget to oversampling in partially labeled imbalanced streams; and Tao [40] targets high-local-entropy regions adversarially. Closest to our decomposition, Sağlam and Cengiz [24] separate the two uncertainty sources explicitly: their generator flags high-*aleatoric* instances as noise and excludes them from seeding, then directs generation toward high-*epistemic* instances. That design choice is operationally convergent with our findings, that generation seeded in the (aleatoric) overlap region is what harms accuracy, that excluding such seeds removes the harm (clean-masked SMOTE), and that only the epistemic (data-reducible) slice of the deficit benefits from more data. The difference is the object of study: these works propose *new oversamplers* and evaluate them as methods, whereas we supply the instance-level diagnosis and the mechanism that explain *why* aleatoric-region generation hurts (Proposition 1, §5.4), and our remedy decomposition shows the epistemic slice such methods can address is small ($\sim$6% of the minority deficit) next to the threshold-recoverable majority ($\sim$68%), which needs no generation at all.

Label noise detection. Confident Learning [41] estimates the joint distribution of noisy and true labels to identify mislabeled instances. Our Cat 1 (noise) category captures a similar signal but is embedded in a broader three-way decomposition that distinguishes noise from boundary ambiguity — a distinction Confident Learning does not make.

Uncertainty decomposition. Shaker and Hüllermeier [10] decompose random-forest predictive uncertainty into aleatoric and epistemic components via ensemble-of-trees disagreement. Hüllermeier and Waegeman [11] provide a general treatment of the aleatoric–epistemic distinction, and Nguyen et al.

[42] use it for data acquisition (epistemic uncertainty marks the *reducible* part an extra label can resolve, aleatoric the irreducible part) in an active-learning setting. We adapt this reducible-versus-irreducible logic to imbalanced learning: combined with a local class-ratio measure and a threshold-recoverability test, it yields the remedy-mapped decomposition of the misclassified minority that is our headline contribution.

Recent reviews. Fernández et al. [3] mark the 15-year anniversary of SMOTE with a progress-and-challenges review. Carvalho et al. [6] survey resampling approaches from a data perspective; their review reinforces the observation that resampler performance is best predicted by data structure rather than by the algorithmic family of the resampler, a perspective consistent with our exclusion-beats-targeting finding.

3. Method

Figure 1 gives the end-to-end pipeline: a class-weighted out-of-bag forest ensemble supplies the model-derived signals; misclassified instances are triaged into three categories by out-of-bag confidence and local-representation rules whose interpretation is grounded in, and validated against, an aleatoric–epistemic uncertainty decomposition (§3.1–3.2); the misclassified *minority* is then mapped to remedies by the remedy decomposition (§3.3); and the resulting diagnosis is evaluated against a 16-strategy menu (§3.5, §4). Table 1 states the procedure as pseudocode.

3.1. Uncertainty decomposition via random forest ensembles

We train an ensemble of $M = 5$ random forests [43], each with $T = 100$ trees and minimum leaf size 5, on the full training set. For each instance $\mathbf{x}_i$, each tree t in forest m produces a class probability vector $\hat{p}_{m,t}(\mathbf{x}_i)$. Following Shaker and Hüllermeier [10, 11], we decompose the total predictive uncertainty:

$$\underbrace{H[\bar{p}(\mathbf{x}_i)]}_{\text{total}} = \underbrace{\frac{1}{MT}\sum_{m,t} H[\hat{p}_{m,t}(\mathbf{x}_i)]}_{\text{aleatoric}} + \underbrace{H[\bar{p}(\mathbf{x}_i)] - \frac{1}{MT}\sum_{m,t} H[\hat{p}_{m,t}(\mathbf{x}_i)]}_{\text{epistemic}} \qquad (1)$$

where $\bar{p}(\mathbf{x}_i) = \frac{1}{MT}\sum_{m,t}\hat{p}_{m,t}(\mathbf{x}_i)$ is the ensemble-averaged prediction and $H[\cdot]$ denotes Shannon entropy. The ensemble size follows standard deep-ensemble practice (M=5, [44]); a sensitivity analysis over $M \in \{1, 2, 5, 10, 20\}$ shows

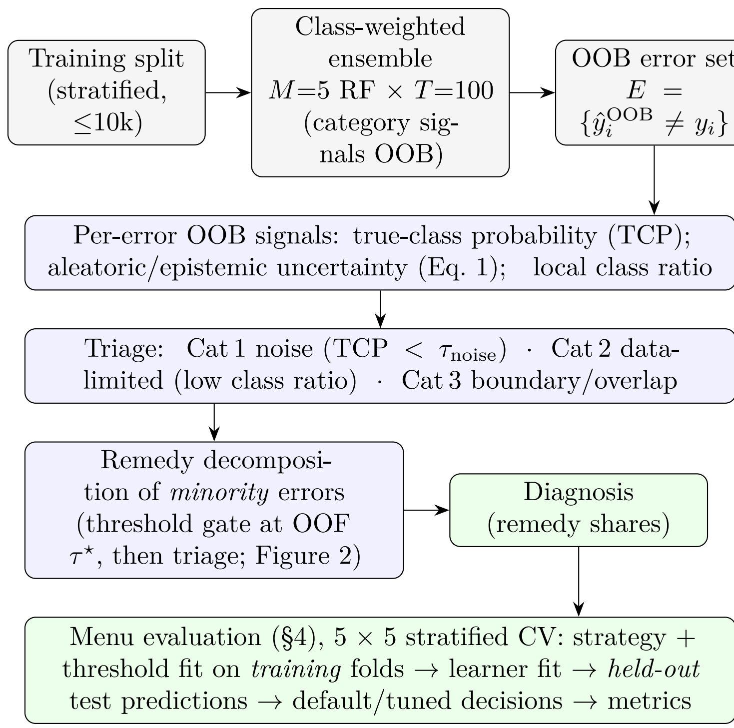

Figure 1: End-to-end pipeline. Every signal that drives the categorization (error set, true-class probability, noise consensus) comes from out-of-bag predictions of a class-weighted random-forest ensemble (the uncertainty values of Eq. (1) interpret the categories and are recomputed fully held-out in Appendix B); errors are triaged into three categories; the misclassified minority is mapped to remedies via the threshold-recoverability gate of §3.3. The diagnosis is then evaluated against the 16-strategy menu under the protocol of §4: 5 × 5 stratified cross-validation in which each strategy (resampling, weighting, or threshold-moving) and any tuned threshold are fit on the training folds only, and all reported metrics come from the held-out test folds.

the resulting categorization and remedy shares are stable for $M \geq 5$ (category fractions move ≤ 2.6 pp between M=5 and M=20; Appendix G), and Appendix F re-derives the full decomposition with bagged non-tree ensembles (MLPs, logistic regression), with the headline unchanged.

The decomposition separates two distinct sources of error. Aleatoric uncertainty captures the irreducibility of the data-generating process: it is high in regions of genuine class overlap. Epistemic uncertainty captures the limits of the current sample: it is high where the model has seen insufficient training data. This distinction matters for rebalancing. Boundary-overlap errors (high aleatoric) are not correctable by adding more data of the existing class; sparse-region errors (high epistemic) are exactly where additional same-class evidence helps.

By contrast, the k-nearest-neighbour categorizations of Napierała & Stefanowski [15] and related work [16] measure only local same-class geometry, which conflates these two sources: an instance with 0 same-class neighbours might be a labelling error, a Bayes-boundary point in a high-overlap region, or a genuinely under-represented sparse instance. We use the aleatoric–epistemic decomposition purely as a *diagnostic* (to localize the boundary/overlap region on real data and characterize the error population, §5), not as a new oversampler.

Out-of-bag estimation (no in-sample optimism). Every *model-derived* signal that drives the categorization is computed from *out-of-bag* (OOB) predictions, i.e. for each instance, only from the trees that did not bootstrap it. An instance is an *error* if the OOB ensemble majority vote disagrees with its label; the true-class probability (TCP, §3.2) is the OOB estimate; the noise signals use OOB confidence and consensus. The remaining signals (the local class ratio) are properties of the data geometry, not model predictions. The error set and its categories are therefore *not* in-sample artefacts of forests fit and scored on the same data. One quantity is scoped differently: the aleatoric/epistemic values of Eq. (1) are computed over all trees of the fitted instrument (in-bag included); they do not enter the category rules of §3.2, serving as interpretation and validation, and the strict check below recomputes them fully held-out. As a strict check we additionally re-derive the categories under inner cross-validation (forests fit on inner-training folds categorize held-out instances, so even the aleatoric/epistemic decomposition is out-of-fold); the category distribution and the aleatoric/epistemic separation are essentially unchanged (Appendix B). We then categorize each error instance.

3.2. Three-way error categorization

For each error instance $\mathbf{x}_i$ we compute a *local class ratio*: the fraction of k-nearest neighbours sharing its label, normalized by the global class frequency. Low values indicate the class is locally under-represented (a data gap); high values indicate adequate local representation despite the error (genuine overlap). The categorization rules are:

Cat 1 (Noise): $\mathrm{TCP}(\mathbf{x}_i) < \tau_{\mathrm{noise}}$, where TCP is the ensemble true-class probability. Very low TCP indicates likely mislabeling; $\tau_{\mathrm{noise}} = 0.12$ gates noise from sparse-boundary errors.

Cat 2 (Data-limited): not Cat 1, and $\mathrm{class\_ratio}(\mathbf{x}_i) < \tau_{\mathrm{cat2}}$ ($\tau_{\mathrm{cat2}} = 0.4$). The class is locally under-represented; the error arises from insufficient same-class data, not overlap.

Cat 3 (Irreducible): neither Cat 1 nor Cat 2. The class is adequately represented locally yet the model errs, indicating genuine class overlap in a model-estimated Bayes-boundary/overlap region. (We reserve the term “Bayes boundary” for the controlled experiments of §5.3 where it is known; on real data Cat 3 is an *estimate* of that region.)

The rules are deliberately operational — ensemble confidence (TCP) and local representation — rather than thresholds on the uncertainty values of Eq. (1) themselves; the aleatoric–epistemic decomposition supplies the *interpretation* of the resulting categories, and the held-out validation confirms it (Cat 3 aleatoric uncertainty exceeds Cat 2’s on 80% of datasets, Appendix B).

To remove the imbalance bias of an off-the-shelf ensemble (a majority-biased ensemble suppresses minority TCP and over-flags minority instances as noise), the triage ensemble is trained with class-balanced instance weights (inverse class frequency). Forest probability estimates respond only weakly to such weights (§5.5), and the *unweighted* bagged instruments of Appendix F reproduce the decomposition, so the headline does not hinge on this choice.

The two thresholds are fixed at $\tau_{\mathrm{noise}} = 0.12$, $\tau_{\mathrm{cat2}} = 0.4$ across *all* datasets; they are defaults, not tuned hyperparameters. The categorization is robust to their exact values: sweeping $\tau_{\mathrm{noise}} \in [0.05, 0.30]$ and $\tau_{\mathrm{cat2}} \in [0.2, 0.6]$, 74% of datasets show $< 0.5\%$ downstream-accuracy variation across all configurations, and the category assignments preserve their relative structure throughout (Appendix A).

3.3. Remedy decomposition of the minority deficit

The triage characterizes *why* an instance is hard; our central instrument turns this into *which remedy applies* to the instances that actually drive the deficit — the misclassified minority. Two instrument choices matter here and we state them explicitly. First, the deficit being decomposed is that of the *default, off-the-shelf* model, so the decomposition is measured on the *unweighted* ensemble; the class-weighted variant of §3.2 partially enacts the operating-point shift inside the instrument, leaving a harder residue, and is reported as a sensitivity (§5.2). Second, the error set is the ensemble's *actual* default prediction: a minority instance is a default error iff the out-of-bag argmax over its mean probability vector differs from its label (on binary tasks this coincides with $\Pr(\text{minority}) < \frac{1}{2}$; on multiclass tasks the two differ, and the argmax definition is the valid one). For each such error we ask, in order:

Threshold-recoverable: is it classified *correctly* under the decision-rule intervention that predicts the focal minority class whenever $\Pr(\text{minority}) \geq \tau^\star$ and otherwise the highest-scoring remaining class (a one-vs-rest threshold move, valid for binary and multiclass alike), where $\tau^\star$ maximizes minority-vs-rest balanced accuracy? The threshold is *cross-fitted*: instances are split into five stratified folds, and each instance's recovery is judged with a $\tau^\star$ selected on the other folds only, so no recovery decision uses a threshold chosen on that instance. If recovered, the model's score already places it on the minority side of the tuned operating point and only the default cutoff was wrong: the remedy is to change the decision rule, no data and no refitting required. This axis, which conditions on score-recoverability of an already-misclassified instance, is what distinguishes our decomposition from difficulty taxonomies.

Data-reducible (epistemic): otherwise, if the instance is Cat 2 (locally under-represented, high epistemic uncertainty), the remedy is more same-class data there (collection or targeted generation).

Irreducible (aleatoric): otherwise, if it is Cat 3 (model-estimated overlap, high aleatoric uncertainty), no remedy in our menu recovers it. On real data this is an *estimated*-irreducible, aleatoric residue, not a proven Bayes-irreducible one.

(Cat 1 noise is reported separately.) The $\Pr(\text{minority})$ scores used here are out-of-bag with respect to model fitting, and $\tau^\star$ is cross-fitted as above, so the decomposition is held out from both model fitting and threshold selection (Appendix B). This three-way, remedy-mapped split of the *minority error population* is the paper's headline diagnostic; its empirical shape (§5.2) explains why a threshold move recovers most of the deficit and identifies the minority of cases where data genuinely helps.

Remedy-irreducible is not the Cat 3 set. The remedy decomposition is not a relabelling of the triage, and the *irreducible* remedy class is much smaller than the Cat 3 (boundary/overlap) category (Figure 2). Two things separate them. First, the remedy decomposition runs only on the *misclassified minority*, whereas the Cat 3 fraction is reported over *all* errors. Second, and more importantly, it gives threshold-recoverability *priority* over the triage category: an error the triage places in the boundary/overlap region (Cat 3) is still counted *threshold-recoverable* whenever the model ranks it above $\tau^\star$: the default $\frac{1}{2}$ cutoff was wrong, but the *ranking* is right. The irreducible remedy class therefore contains only the Cat 3 minority errors that remain misclassified *after* the threshold move, a strict subset of Cat 3. This is why the irreducible remedy fraction ($\sim$17% of the minority deficit, §5.2) is far below the Cat 3 share of all errors ($\sim$85%, §5.4): most boundary errors are scored well enough to be recovered by moving the decision, not by adding or accepting data.

3.4. Why boundary interpolation crosses the posterior boundary

The mechanism has a simple analytic basis. Let class 1 be the minority, write the class-posterior $\eta(\mathbf{x}) = \Pr(y{=}1 \mid \mathbf{x})$, and let the Bayes boundary be $\mathcal{B} = \{\mathbf{x} : \eta(\mathbf{x}) = \frac{1}{2}\}$ with majority-posterior region $\mathcal{M} = \{\eta < \frac{1}{2}\}$ (where the Bayes-optimal label is the majority). SMOTE forms a synthetic point $\mathbf{z} = \mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)$ with $\lambda \sim \mathcal{U}(0,1)$, $\mathbf{x}_i$ a minority seed and $\mathbf{x}_j$ one of its minority k-nearest neighbours, and labels $\mathbf{z}$ as *minority*.

Proposition 1. *Assume η is continuous along the segment $[\mathbf{x}_i, \mathbf{x}_j]$.*

(i) Suppose a minority seed lies in the overlap region, $\eta(\mathbf{x}_i) < \frac{1}{2}$, and is interpolated with a neighbour $\eta(\mathbf{x}_j) < \frac{1}{2}$. By continuity $\mathcal{M}$ contains an ε-neighbourhood of each endpoint, so every sufficiently short *interpolation step from either endpoint yields $\mathbf{z} \in \mathcal{M}$; if in addition η is quasiconvex along the segment (in particular, approximately affine, as in (ii)), then* every $\mathbf{z} \in [\mathbf{x}_i, \mathbf{x}_j]$ *lies in $\mathcal{M}$. Such a $\mathbf{z}$ carries a minority label in a region the Bayes rule assigns to the majority, i.e. it acts as label noise.*

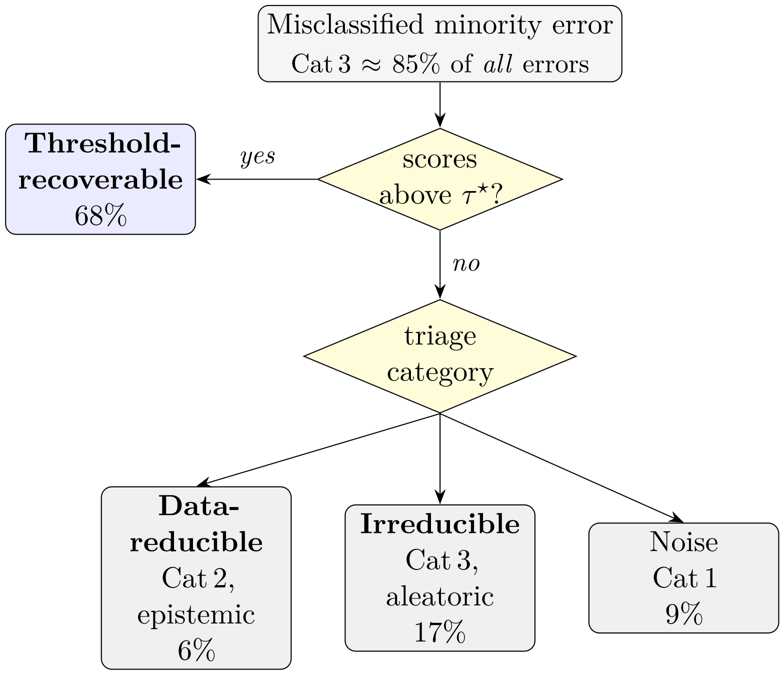

Figure 2: How the misclassified minority filters into the three remedy classes. The decomposition asks threshold-recoverability *first*: an error the model already scores above the cross-fitted threshold $\tau^\star$ is *threshold-recoverable* whatever its triage category. Only the residue is split by triage into data-reducible (Cat 2), irreducible (Cat 3), or noise (Cat 1). Because most boundary (Cat 3) errors are well-scored, the *irreducible* remedy class (17%) is a strict subset of Cat 3 (which is itself ∼85% of *all* errors, §5.4), which is why 17% ≪ 85%. Percentages are means over datasets (Table 3).

(ii) If the segment crosses $\mathcal{B}$, the probability that $\mathbf{z} \in \mathcal{M}$ equals $\left|\{\lambda \in [0,1] : \eta(\mathbf{x}_i + \lambda(\mathbf{x}_j - \mathbf{x}_i)) < \frac{1}{2}\}\right| \in (0,1)$. When η is approximately affine over the segment with $\eta(\mathbf{x}_i) \geq \frac{1}{2} > \eta(\mathbf{x}_j)$, this probability is $\dfrac{\frac{1}{2} - \eta(\mathbf{x}_j)}{\eta(\mathbf{x}_i) - \eta(\mathbf{x}_j)}$.

(iii) Consequently the expected fraction of synthetic minority mass placed in $\mathcal{M}$ is non-decreasing in the local overlap (the minority probability mass near and beyond $\mathcal{B}$) and in the proportion of seeds drawn from the boundary/overlap region.

Proof sketch. (i) $\mathcal{M} = \{\eta < \frac{1}{2}\}$ is the open sublevel set of a continuous function, so it contains an ε-ball around each interior endpoint and sufficiently short interpolation steps stay in $\mathcal{M}$. A sublevel set need not be convex, so the

Table 1: The triage and remedy-decomposition procedure in pseudocode. All model-derived quantities are out-of-bag (OOB) or out-of-fold (OOF); nothing is fit and scored on the same data.

Input: training split $\mathcal{D} = \{(\mathbf{x}_i, y_i)\}$; τ_{noise}=0.12, τ_{cat2}=0.4; M=5, T=100, k
Output: triage categories; remedy shares of the minority deficit

1 Fit M random forests (T trees each, class-balanced instance weights); for each $\mathbf{x}_i$ collect the OOB tree probabilities $\hat{p}_{m,t}(\mathbf{x}_i)$ and their mean $\bar{p}(\mathbf{x}_i)$.
2 Error set $E \leftarrow \{i : \text{OOB majority vote} \neq y_i\}$.
3 For each $i \in E$: $\text{TCP}_i \leftarrow \bar{p}(\mathbf{x}_i)[y_i]$; aleatoric/epistemic uncertainty via Eq. (1); local class ratio $r_i \leftarrow$ fraction of k-nearest neighbours sharing y_i, normalized by the global frequency of y_i.
4 **Triage:** Cat 1 (noise) if $\text{TCP}_i < \tau_{\text{noise}}$; else Cat 2 (data-limited) if $r_i < \tau_{\text{cat2}}$; else Cat 3 (boundary/overlap).
5 **Threshold:** $\tau^\star \leftarrow \arg\max_\tau$ balanced accuracy of the rule $\bar{p}(\text{minority} \mid \mathbf{x}) \geq \tau$ on OOF predictions of the training split.
6 **Remedy decomposition:** for each *minority* $i \in E$: *threshold-recoverable* if $\bar{p}(\text{minority} \mid \mathbf{x}_i) \geq \tau^\star$; else *data-reducible* if Cat 2; *irreducible* if Cat 3; *noise* if Cat 1.
7 Return category fractions over E and remedy shares over the minority errors.

segment is guaranteed to remain in $\mathcal{M}$ *in its entirety* only under the added quasiconvexity (or approximately-affine) hypothesis; absent it, an interior excursion above $\frac{1}{2}$ is possible but does not affect the label-noise conclusion for the points that do fall in $\mathcal{M}$. (ii) follows from the intermediate value theorem and, in the affine case, from solving $\eta(\mathbf{x}_i) + \lambda[\eta(\mathbf{x}_j) - \eta(\mathbf{x}_i)] = \frac{1}{2}$ for the crossing $\lambda^\star$ and integrating the uniform density over $\{\eta < \frac{1}{2}\}$. (iii) follows by taking expectations over the seed/neighbour sampling distribution, whose support shifts toward $\mathcal{B}$ as overlap and boundary-seed proportion grow. □

This explains three observations below: standard SMOTE places an *overlap-dependent* fraction of synthetic points across a *known* boundary (§5.3); augmenting estimated-boundary (Cat 3) seeds reproduces the tradeoff while non-boundary seeds do not (§5.4); and boundary-*targeting* variants (Borderline-SMOTE, ADASYN) raise exactly the boundary-seed proportion of clause (iii), so they do not resolve the tradeoff.

3.5. Strategy menu

We evaluate a menu of 16 rebalancing strategies grouped into five families: (i) *do nothing*: train on the raw data; (ii) *threshold-moving*: choose the decision threshold on out-of-fold training predictions to maximize balanced accuracy (binary tasks), leaving the data untouched; (iii) *cost-sensitive*: class-balanced sample weights [13], and triage-informed weighting (upweighting the correct same-class neighbours of Cat 2 errors); (iv) *generative oversampling*: SMOTE [2] and eight variants (Borderline-SMOTE, ADASYN, Safe-Level-SMOTE, ProWSyn, MWMOTE, polynom-fit, Napierala-guided), plus *clean-masked SMOTE* (SMOTE seeded only from non-boundary, non-noise minority instances, i.e. excluding Cat 1/Cat 3); at review we additionally evaluate the modern geometric variant G-SMOTE [35] under the same protocol, reported alongside the pre-registered menu rather than inside it (§5.3, §5.5, Appendix H); (v) *balanced ensembles*: balanced random forest [45] and EasyEnsemble [46]. The two triage-derived members (clean-masked SMOTE, triage weighting) operationalize the diagnostic; as §5 shows they do not beat the strong non-generative families, and we include them for completeness rather than as headline methods.

3.6. Meta-features and prescriptive selection

To ask whether a dataset's structure determines its best strategy, we describe each dataset by meta-features in three groups: (a) *error-structure* from the triage (Cat 1/2/3 fractions and masses, minority error rate, learnable-minority fraction); (b) *complexity / separability*: Fisher's maximum discriminant ratio (F1) and the boundary/overlap measures N1, N2, N3 [5]; and (c) *basic* descriptors (imbalance ratio, size, dimensionality, class count). A leave-one-dataset-out selector predicts, from these meta-features, which menu strategy maximizes the target metric on the held-out dataset. We evaluate two designs: a per-method regressor (predict each strategy's score, take the argmax) and a direct method-classifier. We compare both against fixed baselines (always-SMOTE, the best single fixed strategy), an imbalance-ratio-only rule, and a per-dataset oracle. Significance is assessed by paired Wilcoxon tests across datasets with bootstrap confidence intervals, and selector stability is checked across random seeds.

4. Experimental Design

Datasets. We evaluate on 79 real classification datasets: the 27 datasets of the `imbalanced-learn`/KEEL benchmark suite (imbalance ratios 8–130) and 52 of the 54 OpenML [47] and scikit-learn datasets listed in Appendix I (USPS and webpage are excluded from the menu evaluation for high dimensionality and sparse formatting), spanning binary and multiclass problems with original sizes from 150 to 581,012 instances and 4 to 500 features. We apply stratified subsampling to a maximum of 10,000 instances per dataset, which preserves the triage category distribution (mean absolute deviation < 2 pp per category, Appendix D). The interventional experiment (§5.4) uses the 50 datasets where minority size permits SMOTE seed-pool slicing. The OpenML roster is listed in Appendix I.

Strategy menu and base learner. We evaluate the 16-strategy menu of §3.5. The base learner for non-ensemble strategies is XGBoost [48] (300 trees, max depth 6, learning rate 0.1); balanced random forest (300 trees) and EasyEnsemble (10 AdaBoost members) are themselves ensemble learners. We use 5×5 Repeated Stratified K-Fold cross-validation; the triage and the moved threshold are fit on the training split only (threshold chosen on out-of-fold training predictions, so there is no test leakage). Binary-only strategies (threshold-moving, EasyEnsemble) are skipped on multiclass datasets and excluded from those datasets' candidate sets.

Metrics. Balanced accuracy is the primary metric (it is the objective that motivates rebalancing), and we report accuracy alongside it to expose the tradeoff, plus G-mean and MCC as tradeoff-internalizing checks.

Statistical protocol. Because raw scores are heterogeneous across datasets, we never pool them; all cross-dataset numbers are either (a) *paired per-dataset differences* between two strategies on the same dataset (reported as mean/median percentage-point deltas with win rates and Wilcoxon signed-rank [49] p-values), or (b) *average ranks* across strategies, compared with a Friedman test and Nemenyi post-hoc analysis [50] (the critical-difference diagram of §5.5). Pairwise contrasts therefore use Wilcoxon; menu-wide rankings use Friedman–Nemenyi. Per-dataset values are means over the 25 folds; raw per-dataset scores are used only to compare strategies within the same dataset.

Selection protocol. The prescriptive selector (§3.6) is evaluated leave-one-dataset-out across the 79 datasets: for each held-out dataset the selector is trained on the other 78 and its chosen strategy's realized metric is recorded. We compare against a per-dataset oracle, the best single fixed strategy, always-SMOTE, an imbalance-ratio-only rule, and random choice, with paired Wilcoxon tests, bootstrap confidence intervals on the mean gain, and a seed-robustness check over 8 random seeds.

Pre-registration. To avoid post-hoc selection of the analysis, we pre-specified the primary selector (a per-method regressor over all meta-features) and the primary metric (balanced accuracy) before running the full-menu evaluation. The direct method-classifier and the parsimonious feature set are reported as secondary analyses, and we disclose in full that the pre-registered regressor only ties the best fixed strategy while the classifier modestly beats it (§5.7).

5. Results

We proceed from validating the triage, to the headline remedy decomposition (the deficit is mostly threshold-recoverable), to the mechanism (controlled overlap and a proposition), to its consequence at scale (oversampling is an operating-point move: no ranking gain, a prior shift in probability space rather than a calibration change, and a tie with threshold-moving at parity), and finally to the prescription (a regime map and an honestly-scoped selector).

5.1. Error Instance Triage: validation

Synthetic ground truth. On two-class Gaussian mixtures the triage behaves as designed. As class separation shrinks from 1.2 to 0.2 standard deviations (overlap grows), the Cat 3 (irreducible) fraction among errors rises monotonically, exceeding 95% at the smallest separation (mean 94.3% across the sweep). The triage correctly attributes boundary-overlap errors to irreducibility. On 24 real datasets with 5% injected label noise the triage detects the flips with mean precision 0.77, and the flagged-noise fraction rises significantly ($p = 2 \times 10^{-5}$, Wilcoxon). The triage is thus sensitive to both boundary overlap (Cat 3) and label noise (Cat 1), the two categories that drive the mechanism below.

Out-of-fold robustness. Because the categories underpin the mechanistic argument, we verify they are not in-sample artefacts. The category assignment already uses out-of-bag predictions (§3.1); under a strict inner cross-validation that re-derives *every* signal (including the aleatoric/epistemic decomposition) on held-out instances, the Cat 3 fraction among errors is essentially unchanged (in-sample $\rightarrow$ out-of-fold mean shift of ≈ 1.9 pp over 79 datasets) and the aleatoric/epistemic ordering that defines the categories is preserved (Appendix B).

Is the triage a uniquely good predictor of SMOTE harm? No, and it need not be. It is reasonable to ask whether the aleatoric–epistemic triage is necessary or whether a simpler k-NN hardness / overlap score localizes SMOTE's harm equally well. Across the 75 datasets with a SMOTE outcome we correlate eleven per-dataset measures with SMOTE's accuracy cost (Table 2). The trivial *minority error rate* is the strongest single predictor ($\rho = -0.49$, leave-one-out $R^2 = 0.34$), with the Napierała rare/outlier fraction and the imbalance ratio close behind ($\rho \approx -0.40$); the triage's Cat 3 mass is only a moderate, marginally significant predictor ($\rho = -0.21$, $p = 0.07$). The *conditional* Cat 3 fraction (`frac_cat3`) even correlates with *less* accuracy harm ($\rho = +0.37$): the fraction of errors that are boundary errors is not the dataset-level *mass* of boundary-entangled minority, and a dataset can carry a high Cat 3 share within a *small* error population, a near-ceiling problem SMOTE barely perturbs. The mechanism is tied to that boundary mass (better captured by the minority error rate), not the conditional fraction alone. We state this plainly: the triage is *not* a better cross-dataset forecaster of SMOTE harm than simple measures, and we do not claim it is. Its role is *explanatory* (it localizes *which* errors lie in the boundary/overlap region so the intervention of §5.4 can seed augmentation there and probe the mechanism), not a predictive instrument, consistent with the honest selection finding of §5.7. Put differently, the minority error rate predicts the *size* of the deficit; the remedy decomposition diagnoses the *appropriate intervention*.

5.2. The minority deficit is mostly threshold-recoverable

Result. We apply the remedy decomposition (§3.3) to the misclassified minority on every dataset (Table 3). Across the 67 datasets with a non-trivial minority error set (≥ 5 default errors), the deficit is **predominantly threshold-recoverable**: a mean of **68**% (median 75%) of minority errors are corrected by the cross-fitted threshold move alone: these instances' scores already sit above the tuned operating point. Only 6% are data-reducible (epistemic, where more same-class data would help), 17% are irreducible (aleatoric, model-estimated overlap), and 9% are noise; threshold-recoverable is the largest remedy class on 84% of datasets. The pattern *sharpens* with imbalance: on IR > 3 datasets 80% of the deficit is threshold-recoverable. This is corroborated independently by separability: on the 30 hardest binary datasets (baseline balanced accuracy < 0.85) the baseline classifier's out-of-fold AUC is a median of 0.91 (93% exceed 0.70). The classes are well ranked

Table 2: Triage-design ablation: Spearman correlation of each per-dataset measure with SMOTE's accuracy change (Δacc) and balanced-accuracy change (Δbacc), and a single-feature leave-one-out R^2 for Δacc, over 75 datasets. The triage's Cat 3 mass is not a stronger predictor of SMOTE harm than simple hardness/imbalance measures; the triage's value is explanatory, not predictive. *: $p < 0.05$.

Per-dataset measure	ρ(SMOTE Δacc)	ρ(Δbacc)	LOO R^2(Δacc)
minority error rate	-0.49*	+0.57	0.34
Napierala rare+outlier	-0.40*	+0.58	0.17
imbalance ratio	-0.40*	+0.76	-0.23
frac_cat3 (triage)	+0.37*	-0.53	-0.00
LOO 1-NN error (N3)	-0.24*	-0.06	-0.01
cat3_mass (triage)	-0.21	-0.12	-0.01
boundary frac (N1)	-0.20	-0.15	-0.02
Fisher F1	-0.13	+0.45	0.01
instance hardness (max_fdr)	+0.13	-0.45	-0.04
intra/extra ratio (N2)	+0.06	+0.34	-0.03
Napierala borderline	+0.02	+0.04	-0.03

even where default-threshold balanced accuracy is poor. (The genuine exceptions, AUC $\approx$ 0.59, are the near-irreducible problems.)

Instrument sensitivity. Measuring the decomposition on the *class-weighted* ensemble of §3.2 instead lowers the threshold-recoverable share to 57% (67% at IR $>$ 3), for the reason §3.3 anticipates: the weights pre-enact part of the operating-point shift inside the instrument, so its surviving default errors are the post-shift residue. Threshold-recoverable remains the largest remedy class under both instruments. Standardizing features before the k-NN geometry signals leaves the decomposition essentially unchanged (57.5% vs 57.4% on the weighted instrument), addressing the unit-dependence concern of §6 limitation (v). The result is also not an artefact of the random-forest instrument: re-deriving the full decomposition with bagged MLP and bagged logistic-regression ensembles under identical definitions keeps threshold-recoverable the dominant remedy class at 62% and 63% respectively (73% at IR $>$ 3 for both), with per-dataset correlations of $r = 0.78$ and 0.72 against the forest instrument (Appendix F).

Table 3: Remedy decomposition of the misclassified minority (mean over the 67 datasets of the 79-dataset roster with ≥ 5 minority default errors). Default errors are the ensemble's out-of-bag argmax mistakes; recovery is judged under a cross-fitted threshold (§3.3); the instrument is the unweighted ensemble (class-weighted sensitivity in the text). Columns: fraction of minority errors that are threshold-recoverable (the decision-rule move corrects them), data-reducible (epistemic; more minority data helps), irreducible (aleatoric; model-estimated overlap), or noise. The deficit is predominantly a thresholding artefact, increasingly so with imbalance.

Subset	threshold-recoverable	data-reducible	irreducible	noise
All datasets	68%	6%	17%	9%
IR > 3	80%	4%	5%	12%
IR > 10	80%	5%	2%	13%
Binary	69%	4%	15%	12%
Multiclass	67%	10%	22%	1%

Consistency with Cat 3 dominating the errors. The ∼85% Cat 3 share reported below (§5.4) and the 17% irreducible fraction here are not in tension (§3.3). Cat 3 is a triage label for errors in the estimated overlap region under the *default* decision rule, computed over *all* errors; the remedy decomposition then asks, for the *minority* errors only, whether that default-threshold mistake survives a move to the tuned operating point $\tau^\star$. Because the classifier scores most of these boundary errors above $\tau^\star$, they are counted threshold-recoverable, and only the residue the threshold move cannot rescue is counted irreducible, a strict subset of the minority Cat 3 errors, not the whole category.

Why this matters. This decomposition is the mechanistic account of a well-documented but unexplained regularity, that resampling does not improve ranking and a threshold move reproduces its benefit [12, 13, 9, 14]. *Because* most of the minority deficit is threshold-recoverable, a non-generative change of the decision rule recovers it, and SMOTE, which leaves ranking unchanged (§5.5), is at best an implicit, inefficient way to enact the same threshold move. The decomposition also tells a practitioner *which* situation they are in: the (minority) data-reducible fraction is where collecting or generating minority data is warranted; the irreducible fraction is where no remedy in our evaluated menu helps.

5.3. Mechanism: SMOTE generates across the Bayes boundary

Premise. If SMOTE's accuracy cost arises from generation near the Bayes boundary, then on a controlled problem with a *known* boundary, standard SMOTE should place synthetic minority points *across* that boundary (in majority territory, where they act as label noise), increasingly so as overlap grows, while excluding the irreducible seeds should prevent it.

Result. We generate 2D Gaussian classes ($\mathcal{N}([\pm s, 0], I)$, 9:1 imbalance, balanced Bayes boundary $x = 0$) and sweep the separation s. Standard SMOTE places an *overlap-dependent* fraction of its synthetic minority points on the majority side of the boundary (from 6% at $s = 1.5$ to 34% at $s = 0.4$), whereas masking the Cat 3 (boundary) seeds generates essentially none ($\leq$ 2%, 0% for $s \geq 0.8$); Figure 3 visualizes both.

Boundary-targeting and geometric variants. The same measurement answers whether the mechanism is specific to SMOTE's linear interpolation. As Proposition 1(iii) predicts, the boundary-targeting variants cross *more*: ADASYN places the largest wrong-side fraction at every separation (24–37%, still 24% at the widest separation, because its sparse-seed weighting concentrates on the overlap stragglers), with Borderline-SMOTE between it and SMOTE at wide separations. Geometric SMOTE [35] is the instructive case. Its hypersphere is bounded by the nearest *majority* neighbour, a design-level guard against generating in the other class's territory, and we expected it to cross less. It does not: its wrong-side fraction matches standard SMOTE's at every separation (34% vs 34% at s=0.4; 8% vs 6% at s=1.5). The guard bounds how far generation extends *past* the nearest majority point, but a seed that already lies in the overlap region is already across the boundary, so any point generated around it carries a minority label in majority-posterior territory, exactly clause (i) of the proposition. What drives the harmful mass is *where the seeds sit*, not the geometry of the interpolation, which is why seed exclusion (clean-masking) is the only intervention that removes the crossing.

5.4. Interventional evidence

Premise. If the accuracy–balanced-accuracy tradeoff is *caused* by generation near the (estimated) boundary, then targeted generation around Cat 3 (irreducible) errors should reproduce the SMOTE signature, while random-instance generation should not.

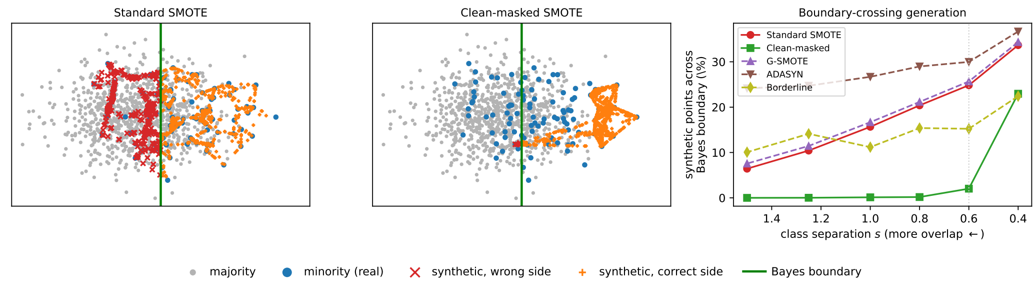

Figure 3: Mechanism on controlled 2D Gaussians (9:1; balanced Bayes boundary $x = 0$, green). Left/middle: at moderate overlap ($s = 0.6$) standard SMOTE scatters synthetic minority points (red $\times$) across the boundary into majority territory; masking the boundary seeds keeps generation on the minority side. Right: the fraction of synthetic points crossing the boundary rises with overlap for SMOTE, for the boundary-targeting variants (ADASYN highest at every separation), and for the geometrically guarded G-SMOTE, whose curve matches standard SMOTE's; it stays near zero only when boundary seeds are excluded.

Protocol. On each training fold we compute the triage and, for each strategy, append a *fixed* budget B of synthetic points equal to the total number of triage errors on that fold, identical across strategies, so the comparison isolates *where* one generates, not how much. Synthetic points are produced by SMOTE-style interpolation *within the targeted seed subset only* (each synthetic point takes its seed's own class, with B split across classes in proportion to the seed subset's class distribution); all original training instances are kept, and the data are *not* rebalanced to a fixed ratio. The six strategies are: `augment_cat2` and `augment_cat3` (seeds from the respective error category), `augment_all_errors` (all errors), `augment_random` (a size-B random sample of *all* training instances, class-proportional, hence a true null that cannot shift the class balance), and two removal controls (`remove_noise`, `remove_random`). The base learner is XGBoost; metrics are over 5×5 folds. Seeds come from the triage on the training fold. This real-data intervention is a *small* effect and, as we report honestly, sensitive to seed re-estimation (Appendix B); we treat it as corroboration of, not primary evidence for, the mechanism, which rests on the controlled experiment (§5.3, known boundary) and the discrimination/threshold analysis (§5.5).

Result. We run these strategies on each of 50 datasets against a no-augmentation baseline. Augmenting Cat 3 errors recreates the signature tradeoff: balanced accuracy improves (win rate 68%, $p = 0.001$) while accuracy trends negative.

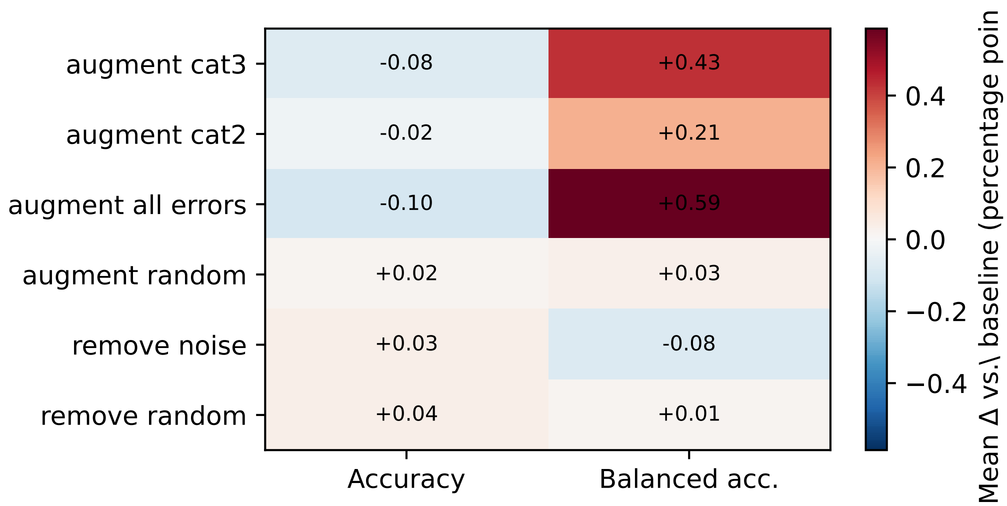

Figure 4: Interventional validation: mean metric change vs. the no-augmentation baseline by augmentation target (50 datasets, fixed budget). Cat 3 (boundary) augmentation reproduces the SMOTE signature (balanced-accuracy gain at an accuracy cost), while random augmentation is null.

Augmenting all errors amplifies it ($p < 0.001$) because Cat 3 dominates the error population ($\sim$86%). Crucially, *random* augmentation produces no effect on either metric (WR $\approx$ 50%, $p > 0.5$), and augmenting Cat 2 errors helps balanced accuracy modestly ($p = 0.007$) without harming accuracy (Figure 4).

Interpretation. Consistent with the mechanism, the targeted-augmentation contrast points to generation in the model-estimated boundary/overlap region (Cat 3) as the source of the accuracy cost, with the random-augmentation control null. We are deliberately measured here: this is a *small* real-data effect that manipulates *model-estimated* hard instances (not the true, unobserved Bayes boundary), and it weakens under strict out-of-fold seed re-estimation at reduced power (Appendix B). The mechanism does not rest on it: the controlled experiment of §5.3 (known boundary) and the discrimination/threshold analysis of §5.5 (oversampling does not improve ranking; its gain is a recoverable operating-point shift) are the load-bearing evidence.

5.5. *SMOTE-family oversampling is an operating-point move, not a discrimination gain*

We now confirm the consequence the decomposition predicts, on the full menu and at scale. The results in this section reproduce known findings [12, 9, 14, 19] on a broad roster; we present them as confirmation that grounds the mechanism, not as new claims.

No discrimination gain; a prior shift, not a calibration gain. The SMOTE family does not help the classifier *rank*: on the binary datasets, SMOTE's out-of-fold ROC-AUC (0.911) and PR-AUC (at fixed test prevalence, 0.653) are statistically indistinguishable from the unresampled baseline (0.909, 0.655), and the same holds for G-SMOTE (0.908, 0.649; Table 4). In rank terms across the seven same-learner strategies, the ROC-AUC ranks sit in an indistinguishable 3.2–4.9 band and the baseline attains the best PR-AUC rank (3.4). What oversampling *does* change is the probability scale, and here the verdict is metric-relative, exactly as the critical literature observes (§2): on classical, prevalence-weighted calibration metrics the unresampled baseline is best (Brier 0.028 vs. SMOTE's 0.031; ECE 0.025 vs. 0.030; best average Brier rank 2.6 of 7, Safe-Level-SMOTE worst at 6.0), whereas on *class-balanced* versions of the same metrics (mean of per-class Brier terms; inverse-class-frequency-weighted ECE) the ordering *reverses* (balanced Brier: baseline 0.205, SMOTE 0.169, Safe-Level 0.160), because oversampling inflates the minority-side probabilities that the baseline leaves under-confident. Both verdicts describe the same event: resampling shifts the effective class prior in probability space. That is a *prior shift*, not a calibration improvement — the shift itself is reproduced without data generation by reweighting, the decisions it licenses are reproduced by threshold-moving with the probability scale left intact (Table 5), and it is undone by post-hoc recalibration [19, 14, 23]. Whatever balanced-accuracy SMOTE buys is therefore an operating-point move, not a better classifier.

At threshold parity, oversampling adds little. The headline "win" of non-generative strategies is itself an operating-point artefact. At the *default* threshold, non-generative strategies top the balanced-accuracy ranking on both rosters (best non-gen $\geq$ best gen on balanced accuracy in 25/27 KEEL datasets, +5.76 pp, 95% CI [+3.81, +7.90]; 43/52 OpenML, +2.02 pp, [+1.20, +2.98]; 86% overall; Friedman–Nemenyi Figure 5, χ^2=285.6, $p < 10^{-50}$, threshold-moving best rank 3.15), but only because threshold-moving is tuned and the

SMOTE family is scored at 0.5. When *every* strategy receives the same out-of-fold threshold tuning, almost the entire gap disappears: on the 27 KEEL binary datasets a tuned threshold alone lifts SMOTE's balanced accuracy by +6.4 pp (G-SMOTE's by +8.1 pp), and the best-non-generative − best-generative difference collapses from +5.76 pp at the default threshold to within $\pm$1 pp at parity on the three main learners (Table 5: -0.99 pp for XGBoost, -0.80 for RF, -0.44 for logistic regression). We are precise about the residual: at parity the best *generative* strategy is marginally *ahead* on XGBoost (p=10^{-4}) and RF (p=4×10^{-4}), and not significant by the signed-rank test on logistic regression (p=0.09; the bootstrap CI on the mean, $[-0.93, -0.04]$, marginally excludes zero) or SVM (-0.10 pp, p=0.4); part of this residual is a selection asymmetry (the generative family takes a maximum over five members, the non-generative over four or fewer). The best-vs-best family contrast is a summary statistic; the primary contrasts are the direct pairwise ones above — tuned threshold-moving against tuned SMOTE and tuned G-SMOTE, each a paired per-dataset difference with confidence interval and win rate — which involve no per-dataset selection. The sub-1 pp edge comes with *no* ranking gain and a prior shift rather than a calibration gain (above), against a +6.4 pp threshold effect. G-SMOTE at parity is indistinguishable from plain threshold-moving (-0.08 pp, p=0.79) and from tuned SMOTE (p=0.93). The decisive lever is therefore the *threshold*, not the generative/non-generative choice, which is a fraction of its size. This corroborates, on a 16-strategy menu, the broader-roster conclusion of [9] (73 datasets) and [14] that balancing and threshold-tuning yield equivalent discrimination. Average ranks tell the same story: at the default threshold the do-nothing baseline ranks last or next-to-last of the seven same-learner strategies on every learner (mean balanced-accuracy rank 6.6, 5.9, 6.9 of 7 for XGBoost, RF, and logistic regression), whereas once every strategy is threshold-tuned the ranks compress into a narrow band (2.9–5.4, 3.6–4.5, 3.0–5.2 respectively), with the tuned baseline ranked *first* on logistic regression.

Where the operating-point account ends: undertrained neural learners. Extending the parity analysis to three further learners (Table 5) locates the boundary of this account. Gaussian naive Bayes behaves like the linear case (generative edge -1.63 pp) and RBF-SVM like the tree case (-0.10 pp, p=0.4). The multilayer perceptron is the qualitative exception: at parity the generative family leads by 11 pp, and the reason is visible in the ranking metrics, which the operating-point account requires to be intact. A single early-stopped MLP trained on the imbalanced data *ranks poorly* (mean ROC-AUC 0.769), and SMOTE-family resampling genuinely repairs it (AUC 0.892, +0.123, p=4×10^{-7}, 93% of datasets): for this learner, imbalance corrupts the learned representation itself, so resampling is *not* reducible to a threshold move. (The comparison also carries a structural caveat: MLPClassifier supports no class weighting, so the non-generative family on this axis is threshold-moving alone.) Notably, *bagging* MLPs restores the ranking — the bagged-MLP instrument of Appendix F reproduces the headline threshold-recoverable share — so the exception is specific to single undertrained networks. This is exactly the regime our deep-learning discussion (§6) anticipates, and we state the scope plainly: the operating-point account holds where the classifier's ranking survives imbalance (trees, linear models, naive Bayes, kernel SVMs, bagged ensembles); for single neural networks resampling has genuine representation value beyond the threshold.

Weighted learning is the same lever. A natural objection is that class-weighted (cost-sensitive) learning should make both oversampling and explicit threshold-moving unnecessary. Elkan's equivalence [13] says the three are one lever: reweighting classes, changing class priors by resampling, and moving the decision threshold all shift the same operating point. Our menu contains class-balanced weighting (family (iii), §3.5), and the parity data bear the equivalence out, with an instructive learner-dependence in how *completely* weighting enacts the shift. For logistic regression, weighting alone is the full move: it gains +12.1 pp balanced accuracy over the default-threshold baseline, matches default-threshold SMOTE (+0.25 pp, p=0.16), and subsequent threshold tuning adds essentially nothing (+0.5 pp, p=0.68). For XGBoost the enactment is partial (weighting +4.0 pp, with threshold tuning adding a further +5.6 pp, $p<10^{-6}$), and for random forests it is almost nil (weighting −0.9 pp, p=0.18, with tuning then adding +16.0 pp, $p<10^{-7}$): bagged majority votes respond weakly to sample weights, so the threshold remains the reliable handle. At full parity the three enactments coincide, as the equivalence predicts: tuned weighting sits within ±0.7 pp of both tuned threshold-moving and tuned SMOTE on every learner. Weighting is thus a legitimate non-generative way to move the operating point; like oversampling, however, it shifts the probability scale relative to the untouched baseline (average Brier rank 3.9 vs. 2.6, Table 4), whereas explicit threshold-moving keeps the model's probabilities intact and makes the operating-point choice visible rather than implicit.

Table 4: Probability metrics by strategy (XGBoost, mean over the 27 KEEL binary datasets, out-of-fold). Oversampling does not improve ranking (ROC-AUC, PR-AUC at fixed prevalence) and recall at 10% false-positive rate is unimproved. The calibration verdict is metric-relative: the unresampled baseline is best on the classical, prevalence-weighted Brier/ECE, while the oversampled models are best on their class-balanced versions — two views of the same prior shift (§5.5). The two rank rows are mean per-dataset ranks across the seven strategies (lower is better).

	none	cost	SMOTE	Border	ADASYN	SafeLvl	G-SMOTE
ROC-AUC	0.909	0.905	0.911	0.910	0.910	0.905	0.908
PR-AUC	0.655	0.652	0.653	0.650	0.649	0.632	0.649
Brier	0.028	0.034	0.031	0.031	0.032	0.040	0.031
Brier (bal.)	0.205	0.169	0.169	0.175	0.168	0.160	0.186
ECE	0.025	0.035	0.030	0.029	0.031	0.044	0.030
ECE (bal.)	0.233	0.182	0.185	0.192	0.185	0.170	0.209
Rec@FPR10	0.775	0.766	0.774	0.775	0.773	0.751	0.773
ROC-AUC rank	4.02	4.94	3.65	3.76	3.65	4.74	3.24
Brier rank	2.63	3.93	3.85	4.04	4.59	6.00	2.96

The default-threshold gap scales with imbalance. Because the deficit grows with imbalance, so does the (operating-point) penalty of leaving SMOTE at the default threshold (Table 6): the default-threshold non-generative advantage rises from +4.45 pp at IR $>$ 1.5 to +5.78 pp at IR $>$ 10, and SMOTE's accuracy cost deepens in lockstep ($-0.37 \rightarrow -0.72$ pp) as its balanced-accuracy gain grows ($+2.30 \rightarrow +4.28$ pp). The tradeoff is dose-responsive to imbalance, exactly as the boundary mechanism predicts; the parity result above shows this default-threshold gap is recoverable by a threshold move rather than by generation.

The cost tracks the mechanism. SMOTE's per-dataset accuracy loss scales with the minority error rate ($\rho = -0.50$, $p < 10^{-4}$) and the imbalance ratio ($\rho = -0.41$, $p < 10^{-3}$): it hurts most exactly where the minority is most boundary-entangled and rebalancing is most aggressive — the regime the mechanism predicts.

A modern geometric variant does not escape the tradeoff. Running G-SMOTE [35] under the identical protocol yields the same signature at a weaker dose: on KEEL it trades −0.44 pp accuracy for +1.90 pp balanced accuracy ($p = 1.6\times10^{-4}$; SMOTE: −0.55/+3.88), and on the OpenML roster −0.27 pp accu-

Table 5: Threshold parity: best non-generative − best generative balanced accuracy when *every* strategy is given the same out-of-fold threshold tuning, per base learner (27 KEEL binary datasets; 26 for RF, whose isolet cell is computationally infeasible). On XGBoost, RF, logistic regression, and SVM the difference is within ±1 pp (versus a +5.76 pp non-generative advantage at the default threshold): the default-threshold "dominance" is an operating-point effect. The MLP row is the diagnostic exception, analysed in the text: imbalance degrades a single MLP's *ranking* (AUC 0.769, restored to 0.892 by resampling), so its deficit is not threshold-recoverable; MLPs also support no class weighting, leaving threshold-moving as the sole non-generative entry.

Base learner	n	NG≥G	non-gen advantage	paired p
XGBoost	27	15%	−0.99 [-1.79,-0.45]	0.0001
Random forest	26	19%	−0.80 [-1.42,-0.30]	0.0004
Logistic regression	27	37%	−0.44 [-0.93,-0.04]	0.09
MLP	27	0%	−11.01 [-15.15,-7.07]	$1e-08$
Gaussian naive Bayes	27	30%	−1.63 [-3.73,-0.22]	0.02
SVM (RBF)	27	44%	−0.10 [-0.40,+0.22]	0.4

racy ($p = 0.005$) for a non-significant +0.47 pp balanced accuracy (SMOTE: −0.33/+1.45); its balanced accuracy is significantly *below* SMOTE's (Appendix H). Consistent with its unchanged boundary-crossing per synthetic point (§5.3), the geometric redesign buys a smaller operating-point move, not an escape from the mechanism: it still pays accuracy for recall while a tuned threshold obtains the shift with no synthetic data at all.

Not a base-learner artefact. The pattern is not specific to gradient-boosted trees. With random-forest and logistic-regression base learners the default-threshold non-generative advantage persists (+2.07 pp, 75%, $p = 1.6 \times 10^{-4}$ for RF; +6.44 pp, 96%, $p = 2 \times 10^{-10}$ for logistic regression, the most miscalibrated learner), and the threshold-parity analysis (Table 5) likewise holds across all three learners, confirming the gap is an operating-point effect rather than a property of any one model family.

5.6. A regime map: which strategy a dataset needs

Result. Aggregating the per-dataset oracle choice into five strategy families, the best move is a balanced ensemble on 42/79 datasets, threshold-moving on 17, cost-sensitive learning on 6, and *oversampling on only* 12 (15%); doing nothing is best on 2. Relative to do-nothing the mean balanced-accuracy advantage is +8.1 pp for threshold-moving, +6.1 for balanced ensembles, +3.3

Table 6: The default-threshold non-generative advantage and the SMOTE tradeoff by data stratum (existing benchmark cells, no re-run), all at the *default* threshold. NG≥G is the fraction of datasets where the best non-generative strategy matches/beats the best generative one on balanced accuracy; "bacc adv." is the mean per-dataset advantage; the last two columns are SMOTE's mean accuracy and balanced-accuracy change vs. the do-nothing baseline (percentage points). The gap grows with imbalance but is an operating-point effect (Table 5).

Subset	n	NG≥G	bacc adv.	SMOTE Δacc	SMOTE Δbacc
IR > 1.5	53	87%	+4.45	-0.56	+3.20
IR > 3	44	89%	+5.27	-0.61	+3.72
IR > 10	32	88%	+5.78	-0.72	+4.28
binary only	46	85%	+4.35	-0.35	+2.96
multiclass only	29	90%	+1.86	-0.39	+1.24
high-overlap	26	77%	+1.68	-0.42	+1.45

for oversampling, and +2.5 for cost-sensitive learning. A depth-3 decision tree over the meta-features (Figure 7) gives an interpretable sketch: low-separability (overlapping) data favours balanced ensembles, while large or well-separated data with a clean minority favours threshold-moving. However, its leave-one-out accuracy (0.53) only matches the predict-majority rate, so the map is best read as *descriptive*: it shows where each family wins, not a rule that beats "always use a balanced ensemble."

5.7. Prescriptive selection: an honest scope

Result. Can per-dataset meta-features prescribe the strategy? We train a leave-one-dataset-out selector on the error-structure and complexity meta-features and compare it to fixed baselines and an oracle (Table 8). The selector beats always-SMOTE decisively (+3.5 pp balanced accuracy, $p < 10^{-4}$), but so does an imbalance-ratio-only rule, because the gain comes from *having non-generative strategies on the menu*, not from sophisticated selection. Over the strongest *fixed* strategy (balanced RF) the edge is real but modest and design-dependent: a direct method-*classifier* recovers ∼60% of the oracle headroom and robustly beats best-fixed on balanced accuracy (+0.64 pp, 8/8 seeds $p \leq 0.01$) and G-mean (+0.64 pp, 8/8 $p \leq 0.03$), but not MCC; a per-method *regressor* and the interpretable rule both tie best-fixed.

Interpretation. This is a direct, held-out, real-data answer to the characteristic-driven selection problem left open by [7]: error-structure features yield a

Table 7: Per-strategy mean accuracy / balanced accuracy across the 16-strategy menu (XGBoost base learner; KEEL, n=27; OpenML, n=52), sorted by mean balanced accuracy. *At the default decision threshold*, non-generative strategies (†) occupy the top of the balanced-accuracy ranking on both rosters and the entire SMOTE family sits below them, but this ranking gap is largely an operating-point effect that collapses once every strategy receives the same out-of-fold threshold tuning (Table 5). Entries are means over the 25 folds; "rank" is the mean per-dataset balanced-accuracy rank over the full menu on the N=46 binary datasets (lower is better; the ranking of Figure 5; the sixteenth member, the balanced clean-masked variant, rank 9.88, is not shown). The significance of the default-threshold advantage (paired Wilcoxon $p < 10^{-10}$, bootstrap 95% CIs) is reported in §5.5.

		KEEL		OpenML		
Strategy	type	acc	bacc	acc	bacc	rank
threshold-moving	†	0.863	**0.856**	0.854	**0.858**	**3.15**
balanced RF	†	0.910	**0.855**	0.812	0.812	5.05
EasyEnsemble	†	0.847	**0.854**	0.815	**0.822**	6.39
cost-sensitive	†	0.958	0.802	0.849	0.793	6.92
SMOTE		0.961	0.800	0.851	0.793	6.64
Safe-Level-SMOTE		0.948	0.806	0.848	0.790	6.46
ADASYN		0.960	0.801	0.851	0.791	7.13
Borderline-SMOTE		0.961	0.795	0.851	0.791	7.68
ProWSyn		0.961	0.800	0.853	0.787	7.14
MWMOTE		0.960	0.795	0.852	0.786	8.71
Napierala-guided SMOTE		0.964	0.783	0.851	0.789	10.04
clean-masked SMOTE		0.964	0.779	0.852	0.786	10.83
polynom-fit SMOTE		0.966	0.766	0.854	0.780	12.78
triage weighting	†	0.967	0.761	0.854	0.778	13.47
baseline (do nothing)	†	0.967	0.761	0.854	0.777	13.72

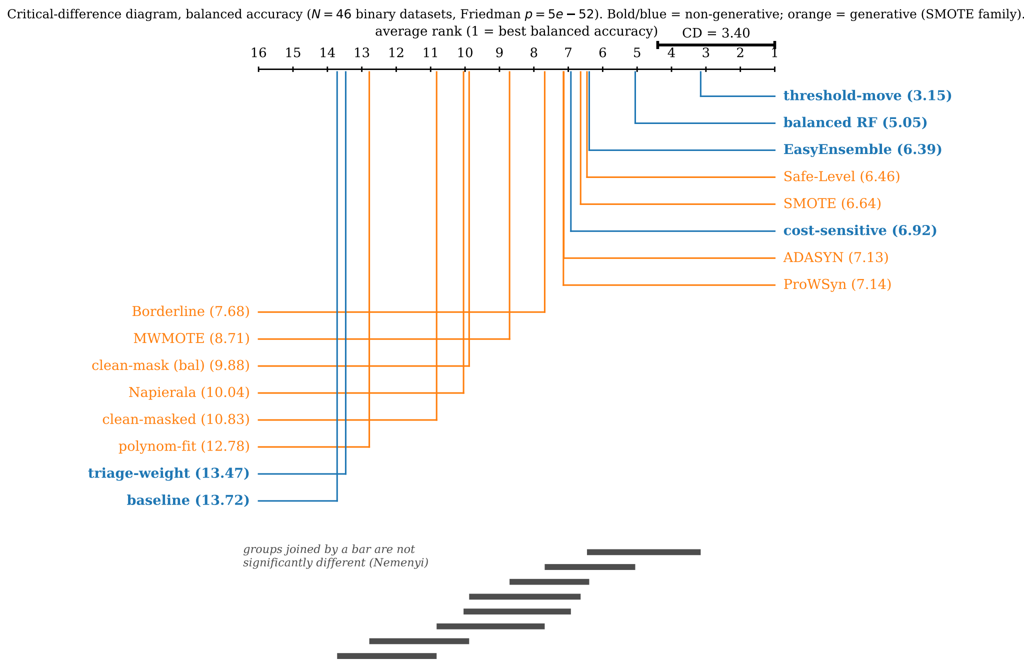

Figure 5: Critical-difference diagram (Friedman–Nemenyi, $\alpha = 0.05$, $N = 46$ binary datasets) of mean per-dataset balanced-accuracy rank over the full 16-strategy menu. Lower rank (right) is better. Non-generative strategies (blue) threshold-moving, balanced RF, and EasyEnsemble attain the best ranks; the SMOTE family ranks in the middle; non-rebalancing methods rank worst.

selector that beats the always-SMOTE paradigm and recovers most of the achievable gain, but the dominant lever is the *menu*, not selector sophistication. Indeed, an imbalance-ratio-only selector matches the full meta-feature selector (Table 8), so the error-structure features add no measurable *selection* value: the triage's role here is explanatory (localizing the boundary for the mechanistic account), not predictive. The actionable recommendation is therefore simple and robust: default to a strong non-generative strategy (balanced random forest, or threshold-moving when calibrated probabilities matter), and reserve oversampling for the minority of regimes where the map says it wins.

5.8. Computational cost

The triage (5 forests $\times$ 100 trees) costs $\sim$1 s on small datasets to $\sim$33 s at $n \approx 50{,}000$, scaling as $O(n \log n)$ in the training-set size (dominated by the

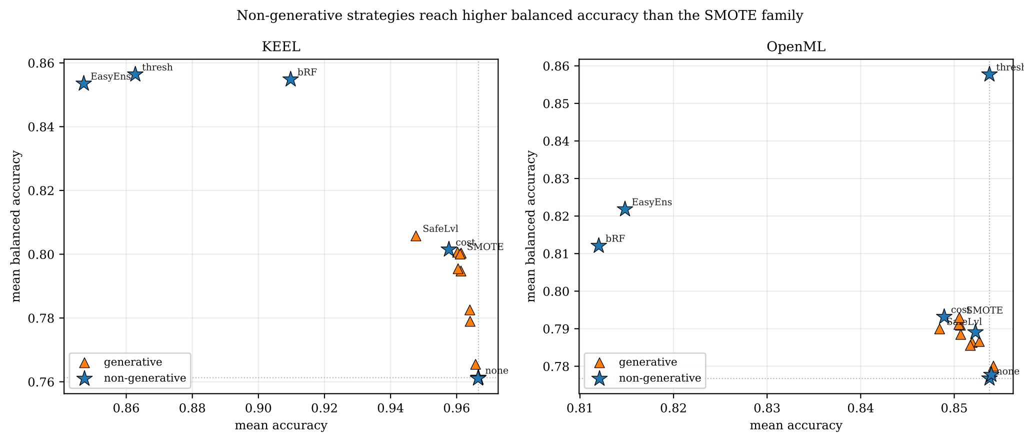

Figure 6: Accuracy–balanced-accuracy plane per roster, *at the default decision threshold*. Non-generative strategies (blue ⋆) reach higher balanced accuracy than the generative SMOTE family (orange △); dotted lines mark the do-nothing baseline. At threshold parity this gap largely closes (Table 5).

forest fits; Appendix E). Crucially it is a *one-time*, classifier-agnostic preprocessing step whose category assignments are reused across every downstream configuration, fold, and base learner, so over a realistic model-selection budget it is a small *amortized* cost, well below the repeated resample-and-refit it informs, not an added per-model overhead. On very large datasets the stratified 10,000-instance subsample (Appendix D) bounds it further without changing the category distribution, so the triage does not become a bottleneck relative to a single SMOTE fit.

6. Discussion

The tradeoff is a decision problem, not a data problem. Our central finding is that the minority-class deficit *decomposes by remedy* and is predominantly *threshold-recoverable*: across 79 datasets a mean of 68% of minority errors (80% at IR > 3) are corrected by moving the decision threshold alone, with only $\sim$6% data-reducible and $\sim$17% irreducible. This explains a regularity the field has documented but not explained, namely that resampling leaves ranking (AUC) unchanged and a threshold shift reproduces its benefit without synthetic generation and without shifting the probability scale [12, 9, 14, 19], and reframes the long-observed accuracy/recall tradeoff: oversampling is an implicit, inefficient enactment of a threshold move (it shifts

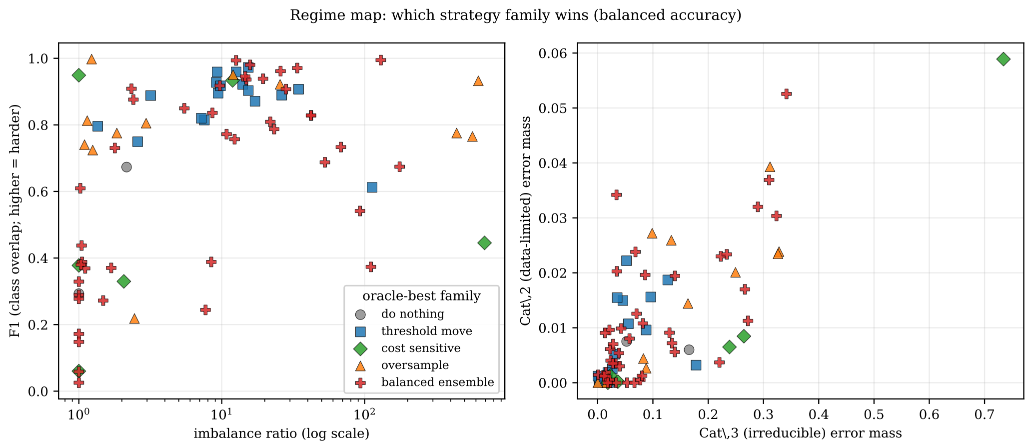

Figure 7: Regime map: the oracle-best strategy family per dataset, over imbalance ratio vs. class overlap (F1, left) and over the triage Cat 3/Cat 2 error masses (right). Oversampling (orange) is best in a minority of regimes; balanced ensembles and threshold-moving cover most datasets.

the operating point by generating across the overlap boundary, at an accuracy and calibration cost), not a way to build a better classifier. Consistent with this, once every strategy is given the same threshold tuning the large default-threshold gap between the families collapses to within ±1 pp.

Exclusion vs. targeting. The dominant paradigm in oversampling research *targets* sparse or boundary regions for generation (ADASYN [27], Borderline-SMOTE [26], and a long tail of variants [4]). We acknowledge their design intent: boundary-targeting variants are often deployed deliberately to buy true-positive rate at some true-negative cost, in applications where missed positives dominate the loss. Our results do not say that operating point is wrong; they say a tuned threshold reaches *any* such recall-biased operating point explicitly, with calibrated probabilities, and without synthetic data (§5.5). Our mechanism explains why boundary targeting has not resolved the accuracy tradeoff itself: these variants all generate at or near the boundary, which is exactly where generation manufactures majority-territory "minority" points. Consistent with this, masking the boundary seeds from SMOTE removes both the accuracy cost *and* the balanced-accuracy benefit (corroborating that the benefit *is* the boundary generation), so the productive move is not a better oversampler but a different lever entirely.

Table 8: Prescriptive selection over the full 16-strategy menu, leave-one-dataset-out across 79 datasets (mean realized metric). A learned selector built on error-structure and complexity meta-features beats always-SMOTE decisively, but over the strongest *fixed* strategy its edge is modest: the direct method-classifier robustly beats best-fixed on balanced accuracy (+0.64 pp, 8/8 seeds $p \le 0.01$) and G-mean (+0.64 pp, 8/8 $p \le 0.03$) but not MCC; the per-method regressor and an imbalance-ratio-only rule both tie best-fixed. The dominant lever is the menu, not the selector.

Selector / fixed strategy	bal. acc	G-mean	MCC
Oracle (per-dataset best, ceiling)	0.838	0.866	0.696
Learned selector (classifier)	**0.833**	**0.862**	0.677
Learned selector (regressor)	0.829	0.858	0.675
Imbalance-ratio-only selector	0.827	0.856	0.673
Best fixed strategy [a]	0.827	0.855	0.680
Always-SMOTE	0.795	0.827	0.678
Baseline (do nothing)	0.772	0.805	0.670

[a] Best fixed strategy is balanced RF for balanced accuracy and G-mean, cost-sensitive for MCC.

The menu matters more than the selector. A recent line of work asks which data characteristics should pick the resampler [7, 6], but its rules barely beat always-SMOTE. We find the reason: given a menu that includes the dominant non-generative strategies, even an imbalance-ratio-only rule beats always-SMOTE, and a learned selector over rich error-structure and complexity meta-features adds only a modest, robust increment over the single best fixed strategy (balanced random forest), significant on balanced accuracy and G-mean, absent on MCC. The practical recommendation is therefore not a sophisticated selector but a change of default: prefer a strong non-generative strategy, and reserve oversampling for the ∼15% of regimes where the map (§5.6) says it wins. Class-weighted learning is an equally legitimate non-generative default where it suits the pipeline, with one caveat our parity analysis quantifies (§5.5): how completely weighting enacts the operating-point shift is learner-dependent (fully for logistic regression, partially for boosting, barely for bagged forests), so explicit threshold-moving remains the more uniformly reliable handle.

Relation to prior categorization and complexity work. Napierała & Stefanowski's instance types [15] and data-complexity measures [5, 25] characterize imbalanced problems descriptively, and meta-learning relates those characteristics to observed performance [6]. Our contribution is orthogonal and mechanism-level: the aleatoric–epistemic decomposition localizes the boundary/overlap region, and the interventional design probes *why* boundary generation hurts, rather than reporting a correlation. This intervention-supported account is what the otherwise extensive selection literature lacks.

Toward deep imbalanced learning. Our base learners are tabular; the analysis is deliberately a controlled, mechanism-level foundation laid before the deep case, and the six-learner parity sweep already contains the first empirical bridge: a single early-stopped MLP is the one learner whose *ranking* imbalance corrupts (AUC 0.769, restored to 0.892 by resampling, §5.5), so for undertrained neural learners resampling has genuine representation value beyond the operating point, while bagging the same MLPs restores the ranking and with it the threshold-recoverable decomposition (Appendix F). Two of the account's three pillars are architecture-agnostic, and we expect them to carry over. The remedy decomposition needs only a ranking score and an aleatoric–epistemic signal, both available from deep models via deep ensembles or MC-dropout [44]; and the threshold-recoverability result is the same operating-point logic that already explains why imbalanced-trained neural networks are systematically miscalibrated and recover under post-hoc threshold/temperature adjustment [51]. The boundary-generation mechanism is the part that needs care: deep oversampling typically interpolates in a *learned latent* space rather than the input space (e.g. DeepSMOTE [52]), so Proposition 1 would apply to the *latent* posterior, whose geometry the encoder itself shapes. We therefore present these results as the necessary tabular proof-of-mechanism and offer a concrete, testable hypothesis: latent-space interpolation across the embedded decision boundary should reproduce the same accuracy/recall tradeoff, and a latent threshold move should likewise recover most of the minority deficit. Testing it across architectures is the natural next thread.

Limitations. (i) The learned selector's edge over a strong fixed default is small ($\sim$0.6 pp) and metric-dependent (present on balanced accuracy and G-mean, absent on MCC); we report it honestly rather than as a headline, and conclude that a fixed non-generative default is the pragmatic recommendation. (ii) The triage thresholds ($\tau_{\text{noise}} = 0.12$, $\tau_{\text{cat2}} = 0.4$) are fixed across datasets; a sensitivity analysis (Appendix A) shows category assignments and downstream accuracy are stable across a wide range. (iii) We validate the default-threshold advantage (and its collapse to within ± 1 pp at threshold parity) across six base learners (gradient-boosted trees, random forests, logistic regression, naive Bayes, RBF-SVM, and MLP; §5.5), so neither the default-threshold gap nor the operating-point explanation is a model-specific artefact; the single MLP is the documented exception (its ranking, not just its threshold, degrades under imbalance), which delimits the account's scope and motivates the deep extension sketched above. (iv) The regime map is descriptive: a depth-3 rule does not beat "always use a balanced ensemble," so the map informs rather than automates the choice. (v) The triage's geometry signals (the local class ratio and the neighbour-based noise signal) use Euclidean k-NN on the loaded feature matrix, with categoricals ordinal-encoded and no standardization; the forest-derived signals are scale-invariant, but the k-NN signals are not, so the Cat 2/Cat 3 boundary can in principle depend on feature units. The threshold sweep of Appendix A covers part of this axis ($\tau_{\mathrm{cat2}} \in [0.2, 0.6]$ leaves downstream conclusions stable), but a fully unit-invariant geometry (standardized or mixed-type distances) is a worthwhile robustness extension.

7. Conclusion

We introduced a remedy decomposition of the misclassified minority class that sorts each error into *threshold-recoverable*, *data-reducible* (epistemic), or *irreducible* (aleatoric), and used it to explain why class-imbalance correction so often reduces to a decision-threshold change. Across 79 real datasets the minority deficit is predominantly threshold-recoverable (mean 68%, 80% at imbalance ratio > 3): the classifier's scores already place these instances above a tuned operating point and only the default cutoff is wrong. This accounts mechanistically for a documented but unexplained regularity, namely that SMOTE-family oversampling does not improve ranking and a threshold move reproduces its benefit without synthetic generation or probability distortion, and recasts the long-observed accuracy/recall tradeoff as an operating-point effect: such oversampling is an implicit, inefficient enactment of a threshold shift, which on controlled data we show it achieves by generating across the overlap boundary. Once every strategy is given the same out-of-fold threshold tuning, the large default-threshold gap between the families collapses to within ± 1 pp across three base learners (six evaluated; naive Bayes shows a small but significant generative edge of 1.6 pp, and the single undertrained MLP, whose ranking imbalance itself corrupts, marks the account's boundary). The decomposition also says *when* the threshold is not enough: the small data-reducible fraction is where minority data genuinely helps, the irreducible fraction where no remedy in our menu does, though a strong non-generative default is near-optimal in the threshold-recoverable majority. The practical message is simple: diagnose the deficit, and for most of it, move the decision rather than oversample.

Appendix A. Threshold Sensitivity

The triage thresholds ($\tau_{\text{noise}} = 0.12$, $\tau_{\text{cat2}} = 0.4$) and the triage-weighting upweight factor ($w = 2.0$, the sample-weight multiplier that the triage-weighting strategy of §3.5 applies to the correct same-class neighbours of Cat 2 errors) are fixed across all datasets. We swept each threshold across a range of values and measured downstream XGBoost accuracy: $\tau_{\text{noise}} \in [0.05, 0.30]$, $\tau_{\text{cat2}} \in [0.2, 0.6]$, $w \in [1.0, 5.0]$. Across the benchmark datasets, 74% show $< 0.5\%$ accuracy variation across all configurations, supporting the use of fixed thresholds as a sensible default. The two methods (clean-masked SMOTE, triage weighting) maintain their relative ordering vs. baselines at every threshold configuration tested.

Appendix B. Out-of-Fold Triage Validation

The triage categories underpin the mechanistic argument (§5.4), so we verify they are not optimistic in-sample artefacts. Two facts establish this. First, by construction every model-derived signal that drives the categorization is *out-of-bag*: the error indicator is the OOB ensemble majority vote, and the true-class probability is the OOB estimate (§3.1); the only non-model signal, the local class ratio, is a property of the data geometry. Second, as a strict check we re-derive the categories under inner 5-fold cross-validation: forests fit on the inner-training folds categorize the held-out instances (via held-out predictions for the error indicator, true-class probability, and the full aleatoric/epistemic decomposition; neighbour-based signals are measured against the inner-training set). Over the 79 datasets, the Cat 3 (irreducible) fraction among errors is essentially unchanged: the in-sample and out-of-fold means coincide at 0.838, the per-dataset Spearman correlation is 0.97 ($p < 10^{-40}$), and the mean absolute shift is 1.9 pp (median 1.0 pp). The aleatoric/epistemic ordering that motivates the categories is preserved out-of-fold (Cat 3 aleatoric uncertainty exceeds Cat 2's on 80% of datasets). The category *identification* is therefore not an in-sample artefact. We note separately that the small real-data *interventional* effect (§5.4) is sensitive to this re-estimation: with out-of-fold seeds and a reduced 3×3 protocol the targeted-augmentation contrast remains directionally consistent (random augmentation null) but is no longer individually significant, which is why we rest the mechanism on the controlled experiment (§5.3) and the discrimination/threshold analysis (§5.5) rather than on the real-data intervention.

Appendix C. GBM Triage Agreement

To verify that the three-way decomposition is not specific to random forests, we replaced the RF ensemble with bagged XGBoost (5 bags of 100 trees, matching the RF configuration). On a 20-dataset agreement subset, the GBM-based triage assigns 82% of training instances to the same category as the RF-based triage. Downstream accuracy under both triages is comparable (mean absolute difference 0.04%), indicating that any ensemble model exposing per-tree predictions can drive the decomposition.

Appendix D. Subsampling Category-Structure Control

We verified that stratified subsampling to 10 000 instances preserves the triage category distribution. On the five largest datasets in our benchmark, we computed the per-category fraction $\{p_{\text{correct}}, p_{\text{noise}}, p_{\text{data-limited}}, p_{\text{irreducible}}\}$ on the full dataset and on a single stratified 10 000-instance subsample. The mean absolute deviation between the two distributions is < 2 percentage points per category across all five datasets, validating the subsampling protocol used throughout (§5.5). Detailed per-dataset numbers are provided in the supplementary material.

Appendix E. Computational Cost Detail

Full cost-table breakdown by dataset (extending §5.8):

The triage time scales roughly as $O(n \log n)$ in the number of training instances and is dominated by the random-forest fits; the local class-ratio computation (k-NN with $k = 5$) contributes $< 5\%$ of total triage time across all dataset sizes.

Table E.9: Triage wall-clock cost vs. a single XGBoost fit (mean over repeats). Overhead = triage time as a percentage of total pipeline time. The triage is a one-time, classifier-agnostic preprocessing cost.

Dataset	n	d	Triage (s)	XGB (s)	Overhead (%)
`iris`	150	4	1.04	0.15	87.2
`wine`	178	13	1.31	0.08	94.1
`sonar`	208	60	1.10	0.12	90.2
`glass`	214	9	1.17	0.96	64.8
`ionosphere`	351	34	1.05	0.12	89.9
`vehicle`	846	18	1.27	0.40	76.1
`vowel`	990	12	1.91	1.59	56.0
`credit-g`	1000	20	1.43	0.16	89.8
`segment`	2310	19	1.56	0.93	63.4
`phoneme`	5404	5	2.71	0.74	77.6
`satimage`	6430	36	2.87	2.19	58.5
`eeg-eye-state`	14980	14	10.20	0.60	94.6
`MagicTelescope`	19020	10	8.40	1.25	88.8
`bank-marketing`	45211	16	16.79	0.84	95.6
`covertype`	50000	54	32.63	49.05	50.1

Appendix F. Instrument Robustness: Non-Tree Ensembles

The remedy decomposition's model-derived signals come from a random-forest ensemble (§3.1). To verify the headline is not an artefact of tree-based uncertainty, we re-derive the full decomposition with two bagged non-tree instruments over the full 27 KEEL + 54 OpenML/scikit-learn roster (81 dataset cells; the headline statistics of Table 3 use the 79-dataset menu roster of §4): B=25 bootstrap-bagged multilayer perceptrons (scaled, 64×32 hidden units, early stopping) and B=25 bootstrap-bagged logistic regressions. Every signal remains out-of-bag: only ensemble members whose bootstrap excluded an instance contribute to its probabilities, and the Shaker-style aleatoric–epistemic decomposition is taken over those members. The local class ratio is the identical k-NN computation, and the error set, one-vs-rest intervention, and cross-fitted threshold follow §3.3 exactly. Two design notes: (i) members are unweighted, matching the headline instrument (§3.3) — class-balanced members enact the operating-point shift *inside* the instrument (completely so for logistic regression, §5.5) and dissolve the default-threshold deficit being

Table F.10: Remedy decomposition by instrument (means over datasets with ≥ 5 minority errors; definitions of §3.3). The threshold-recoverable share, the paper's headline, is stable across instruments; the per-dataset correlation with the RF instrument is given in the last column. The non-tree instruments assign more of the residue to noise (TCP-only gate, note (ii)).

Instrument	thr.-rec.	(median)	data-red.	irred.	noise	r vs RF
Random forests (paper instrument)	68%	(75%)	6%	17%	9%	—
Bagged MLPs	62%	(67%)	3%	16%	19%	0.78
Bagged logistic regression	63%	(68%)	1%	12%	24%	0.72

decomposed; (ii) Cat 1 is gated by TCP alone, as the forest-consensus noise score has no non-tree analogue, which shifts part of the residue from the data-reducible/irreducible classes into noise but leaves the threshold-recoverability axis untouched.

Threshold-recoverable remains the dominant remedy class under both non-tree instruments (Table F.10): 62% for bagged MLPs and 63% for bagged logistic regression against 68% for the RF instrument, rising to 73% on IR > 3 datasets for both (RF: 80%), with per-dataset correlations of $r = 0.78$ and 0.72 against the forest instrument. Part of the gap to the RF share is the TCP-only noise gate of note (ii), which absorbs 19–24% of the residue into noise. The decomposition's ordering, threshold-recoverable far above every other remedy class, is instrument-independent.

Appendix G. Ensemble-Size Sensitivity

The instrument uses $M{=}5$ forests of $T{=}100$ trees, matching standard deep-ensemble practice ($M{=}5$ in [44]). Recomputing the full decomposition for $M \in \{1, 2, 5, 10, 20\}$ on a 10-dataset IR-spanning subset (Table G.11): the triage category fractions move by at most 2.6 pp between $M{=}5$ and $M{=}20$, and the threshold-recoverable share by 3.3 pp on average (1.9 pp for $M{=}10$); the largest single-dataset deviations occur where the minority error set is tiny (a handful of instances, so one flip moves the fraction by many points). Below $M{=}5$ the decomposition is visibly unstable (single-dataset swings up to 34 pp). $M{=}5$ therefore sits at the stability knee, and the conclusions do not change with a larger ensemble.

Table G.11: Ensemble-size sensitivity: mean threshold-recoverable share and absolute deviations from the $M=5$ reference across 10 datasets.

M	mean thr.-rec.	mean $\lvert\Delta$thr.-rec.$\rvert$	max $\lvert\Delta$thr.-rec.$\rvert$	max $\lvert\Delta$cat. fraction$\rvert$
1	81%	5.1 pp	21.9 pp	3.5 pp
2	81%	5.4 pp	34.4 pp	7.4 pp
5 (ref.)	85%	0.0 pp	0.0 pp	0.0 pp
10	84%	1.9 pp	4.7 pp	2.5 pp
20	84%	3.3 pp	6.6 pp	2.6 pp

Appendix H. Extended Oversampler Comparison

Table H.12 reports the full oversampler family, including the Kovács top-ranked variants and Geometric SMOTE (added at review), against standard SMOTE on the OpenML roster.

Table H.12: Oversampler family vs. standard SMOTE (XGBoost, 52 OpenML-roster datasets): mean metric, win rate over SMOTE, and Wilcoxon p. The geometric variant G-SMOTE (added at review) buys significantly *less* balanced accuracy than SMOTE at a comparable accuracy level; only polynom-fit (a known weak-tradeoff variant) and the triage clean-masked variant significantly recover accuracy.

Method (vs. SMOTE)	Mean Acc	WR Acc	p_{Acc}	Mean BAcc	WR BAcc	p_{BAcc}
baseline	0.8538	53.8%	0.001**	0.7768	11.5%	0.000***
borderline_smote	0.8505	38.5%	0.914	0.7911	25.0%	0.013*
adasyn	0.8507	26.9%	0.180	0.7909	32.7%	0.444
safe_level_smote	0.8484	36.5%	0.379	0.7900	34.6%	0.282
polynom_fit_smote	0.8542	55.8%	0.001***	0.7800	17.3%	0.000***
prowsyn	0.8527	48.1%	0.128	0.7866	21.2%	0.000***
mwmote	0.8517	38.5%	0.771	0.7856	28.8%	0.001**
gsmote	0.8510	44.2%	0.677	0.7814	25.0%	0.000***
napierala_guided_smote	0.8507	51.9%	0.103	0.7885	32.7%	0.036*
clean_masked_smote	0.8519	46.2%	0.016*	0.7863	17.3%	0.000***

Appendix I. Dataset Listing

Table I.13 lists the OpenML/scikit-learn roster, including each dataset's OpenML ID, number of instances, features, classes, and imbalance ratio (majority-class count divided by minority-class count); the remaining 27 datasets are the standard `imbalanced-learn`/KEEL benchmark suite.

Table I.13: Full dataset listing (54 datasets; the recovered original-paper roster). n = instances; d = features; C = classes; IR = imbalance ratio (majority / minority count). "sklearn" marks datasets loaded from scikit-learn rather than OpenML; `webpage`'s class statistics are not reported because its sparse high-dimensional format excludes it from the menu evaluation (§4).

Dataset	OpenML ID	n	d	C	IR
`phoneme`	1489	5404	5	2	2.4
`electricity`	151	45312	8	2	1.4
`bank-marketing`	1461	45211	16	2	7.5
`MagicTelescope`	1120	19020	10	2	1.8
`MiniBooNE`	41150	50000	50	2	2.6
`jannis`	41168	50000	54	4	23.2
`covertype`	1596	50000	54	7	109.5
`credit-g`	31	1000	20	2	2.3
`vehicle`	54	846	18	4	1.1
`segment`	36	2310	19	7	1.0
`kc1`	1067	2109	21	2	5.5
`pc4`	1049	1458	37	2	7.2
`ozone-level-8hr`	1487	2534	72	2	14.8
`sick`	38	3772	29	2	15.3
`abalone`	183	4177	8	28	689.0
`yeast`	181	1484	8	10	92.6
`ipums_la_99-small`	1018	8844	56	2	14.6
`mammography`	310	11183	6	2	42.0
`wine_quality`	287	6497	11	7	567.2
`thyroid-allbp`	40474	2800	26	5	52.6

continued on next page

Table I.13 – continued

Dataset	OpenML ID	n	d	C	IR
satimage	182	6430	36	6	2.4
creditcard	1597	50000	29	2	601.4
webpage	350	34780	300	–	–
solar-flare	40687	1066	12	6	7.7
iris	sklearn	150	4	3	1.0
wine	sklearn	178	13	3	1.5
breast_cancer	sklearn	569	30	2	1.7
digits	sklearn	1797	64	10	1.1
glass	41	214	9	6	8.4
vowel	307	990	12	11	1.0
letter	6	20000	16	26	1.1
optdigits	28	5620	64	10	1.0
page-blocks	30	5473	10	5	175.5
sonar	40	208	60	2	1.1
eeg-eye-state	1471	14980	14	2	1.2
waveform	60	5000	40	3	1.0
madelon	1485	2600	500	2	1.0
GesturePhaseSegmentationProcessed	4538	9873	32	5	3.0
JapaneseVowels	375	9961	14	9	2.1
heart-statlog	53	270	13	2	1.2
ionosphere	59	351	34	2	1.8
wine-quality-white	40498	4898	11	7	439.6
wine-quality-red	40691	1599	11	6	68.1
mfeat-factors	12	2000	216	10	1.0
mfeat-fourier	14	2000	76	10	1.0
mfeat-karhunen	16	2000	64	10	1.0
mfeat-morphological	18	2000	6	10	1.0
mfeat-pixel	20	2000	240	10	1.0
mfeat-zernike	22	2000	47	10	1.0
semeion	1501	1593	256	10	1.0
synthetic_control	377	600	60	6	1.0
adult	179	48842	14	2	3.2
splice	46	3190	60	3	2.2
USPS	41082	9298	256	10	2.2

CRediT authorship contribution statement

Cameron Hamilton: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

Declaration of competing interest

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Data availability

All datasets are publicly available (the `imbalanced-learn`/KEEL benchmark suite and OpenML). Code, runner scripts, result files, and a one-command reproduction guide are publicly available at `https://github.com/allianceai/eit`.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used Anthropic's Claude and the Claude Code assistant in order to draft and edit manuscript text, improve language and structure, assist with literature search and LaTeX formatting, and produce figures, diagrams, and tables from the author's own data and analyses. After using these tools, the author reviewed and edited all content, independently verified the results, statistical claims, and cited references, and takes full responsibility for the content of the published article.

References

[1] H. He, E. Garcia, Learning from imbalanced data, IEEE TKDE 21 (9) (2009) 1263–1284.

[2] N. Chawla, K. Bowyer, L. Hall, W. Kegelmeyer, SMOTE: Synthetic minority over-sampling technique, JAIR 16 (2002) 321–357.

[3] A. Fernández, S. Garcia, F. Herrera, N. Chawla, SMOTE for learning from imbalanced data: Progress and challenges, marking the 15-year anniversary, JAIR 61 (2018) 863–905.

[4] G. Kovács, smote-variants: A Python implementation of 85 minority oversampling techniques, Neurocomputing 366 (2019) 352–354.

[5] J. Komorniczak, P. Ksieniewicz, M. Woźniak, Data complexity and classification accuracy correlation in oversampling algorithms, in: LIDTA (PMLR), 2022.

[6] M. Carvalho, A. Pinho, S. Brás, Resampling approaches to handle class imbalance: a review from a data perspective, Journal of Big Data 12 (1) (2025) 71.

[7] Y. Jiang, S. Ye, Beyond imbalance ratio: Data characteristics as critical moderators of oversampling method selection, arXiv preprint arXiv:2604.04541 (2026).

[8] A. S. Tarawneh, A. B. Hassanat, G. A. Altarawneh, A. Almuhaimeed, Stop oversampling for class imbalance learning: A review, IEEE Access 10 (2022) 47643–47660. `doi:10.1109/ACCESS.2022.3169512`.

[9] Y. Elor, H. Averbuch-Elor, To SMOTE, or not to SMOTE?, arXiv preprint arXiv:2201.08528 (2022).

[10] M. Shaker, E. Hüllermeier, Aleatoric and epistemic uncertainty with random forests, in: Advances in Intelligent Data Analysis XVIII (IDA 2020), Vol. 12080 of Lecture Notes in Computer Science, Springer, 2020, pp. 444–456. `doi:10.1007/978-3-030-44584-3_35`.

[11] E. Hüllermeier, W. Waegeman, Aleatoric and epistemic uncertainty in machine learning: An introduction to concepts and methods, Machine Learning 110 (3) (2021) 457–506.

[12] F. Provost, Machine learning from imbalanced data sets 101, in: AAAI Workshop on Learning from Imbalanced Data Sets, 2000, pp. 1–3.

[13] C. Elkan, The foundations of cost-sensitive learning, in: International Joint Conference on Artificial Intelligence (IJCAI), Vol. 17, 2001, pp. 973–978.

[14] R. van den Goorbergh, M. van Smeden, D. Timmerman, B. Van Calster, The harm of class imbalance corrections for risk prediction models: illustration and simulation using logistic regression, Journal of the American Medical Informatics Association (JAMIA) 29 (9) (2022) 1525–1534.

[15] K. Napierała, J. Stefanowski, Types of minority class examples and their influence on learning classifiers from imbalanced data, Journal of Intelligent Information Systems 46 (3) (2016) 563–597.

[16] M. Smith, T. Martinez, C. Giraud-Carrier, An instance level analysis of data complexity, Machine Learning 95 (2) (2014) 225–256.

[17] S. Swayamdipta, R. Schwartz, N. Lourie, Y. Wang, H. Hajishirzi, N. Smith, Y. Choi, Dataset cartography: Mapping and diagnosing datasets with training dynamics, in: EMNLP, 2020, pp. 9275–9293.

[18] N. Seedat, J. Crabbe, I. Bica, M. van der Schaar, Data-IQ: Characterizing subgroups with heterogeneous outcomes in tabular data, in: NeurIPS, 2022.

[19] A. Dal Pozzolo, O. Caelen, R. A. Johnson, G. Bontempi, Calibrating probability with undersampling for unbalanced classification, in: IEEE Symposium Series on Computational Intelligence (SSCI), 2015, pp. 159–166.

[20] K. Napierała, J. Stefanowski, S. Wilk, Learning from imbalanced data in presence of noisy and borderline examples, in: RSCTC, 2010.

[21] J. Sáez, B. Krawczyk, M. Woźniak, Analyzing the oversampling of different classes and types of examples in multi-class imbalanced datasets, Pattern Recognition 57 (2016) 164–178.

[22] C. Drummond, R. C. Holte, C4.5, class imbalance, and cost sensitivity: why under-sampling beats over-sampling, in: ICML Workshop on Learning from Imbalanced Datasets II, 2003.

[23] A. Carriero, K. Luijken, A. de Hond, K. G. M. Moons, B. van Calster, M. van Smeden, The harms of class imbalance corrections for machine learning based prediction models: a simulation study, Statistics in Medicine 44 (3–4) (2025) e10320.

[24] F. Sağlam, M. A. Cengiz, Uncertainty guided oversampling: A classifier-specific approach for effective handling of class imbalance in machine learning, Knowledge-Based Systems 342 (2026) 115886. doi:10.1016/j.knosys.2026.115886.

[25] M. Santos, P. Abreu, N. Japkowicz, A. Fernández, C. Soares, S. Wilk, J. Santos, A unifying view of class overlap and imbalance: Key concepts, multi-view panorama, and open avenues for research, Information Fusion 89 (2023) 228–253.

[26] H. Han, W. Wang, B. Mao, Borderline-SMOTE: A new over-sampling method in imbalanced data sets learning, in: ICIC, 2005, pp. 878–887.

[27] H. He, Y. Bai, E. Garcia, S. Li, ADASYN: Adaptive synthetic sampling approach for imbalanced learning, in: IJCNN, 2008, pp. 1322–1328.

[28] C. Bunkhumpornpat, K. Sinapiromsaran, C. Lursinsap, Safe-level-SMOTE: Safe-level-synthetic minority over-sampling technique for handling the class imbalanced problem, in: PAKDD, 2009, pp. 475–482.

[29] G. Batista, R. Prati, M. Monard, A study of the behavior of several methods for balancing machine learning training data, ACM SIGKDD Explorations Newsletter 6 (1) (2004) 20–29.

[30] D. Wilson, Asymptotic properties of nearest neighbor rules using edited data, IEEE TSMC 2 (3) (1972) 408–421.

[31] I. Tomek, Two modifications of CNN, IEEE TSMC 6 (11) (1976) 769–772.

[32] S. Barua, M. Islam, X. Yao, K. Murase, MWMOTE–majority weighted minority oversampling technique for imbalanced data set learning, IEEE Transactions on Knowledge and Data Engineering 26 (2) (2014) 405–425.

[33] S. Barua, M. Islam, K. Murase, ProWSyn: Proximity weighted synthetic oversampling technique for imbalanced data set learning, in: PAKDD, 2013, pp. 317–328.

[34] S. Gazzah, N. Amara, New oversampling approaches based on polynomial fitting for imbalanced data sets, in: Workshop on Document Analysis Systems, 2008, pp. 677–684.

[35] G. Douzas, F. Bacao, Geometric SMOTE a geometrically enhanced drop-in replacement for SMOTE, Information Sciences 501 (2019) 118–135. doi:10.1016/j.ins.2019.06.007.

[36] A. Sakho, E. Malherbe, E. Scornet, Do we need rebalancing strategies? a theoretical and empirical study around SMOTE and its variants, in: International Conference on Artificial Intelligence and Statistics (AISTATS), 2026, arXiv:2402.03819.

[37] Ş. Ertekin, Adaptive oversampling for imbalanced data classification, in: E. Gelenbe, R. Lent (Eds.), Information Sciences and Systems 2013: Proceedings of the 28th International Symposium on Computer and Information Sciences, Vol. 264 of Lecture Notes in Electrical Engineering, Springer, Cham, 2013, pp. 261–269. doi:10.1007/978-3-319-01604-7_26.

[38] X. Zhang, D. Ma, L. Gan, S. Jiang, G. Agam, CGMOS: Certainty guided minority oversampling, in: Proceedings of the 25th ACM International Conference on Information and Knowledge Management, ACM, 2016, pp. 1623–1631. doi:10.1145/2983323.2983789.

[39] G. J. Aguiar, A. Cano, An active learning budget-based oversampling approach for partially labeled multi-class imbalanced data streams, in: Proceedings of the 38th ACM/SIGAPP Symposium on Applied Computing, ACM, 2023, pp. 382–389. doi:10.1145/3555776.3577624.

[40] L. Tao, Local entropy-adversarial oversampling for imbalanced datasets, Neurocomputing 661 (2026) 131959. doi:10.1016/j.neucom.2025.131959.

[41] C. Northcutt, L. Jiang, I. Chuang, Confident learning: Estimating uncertainty in dataset labels, JAIR 70 (2021) 1373–1411.

[42] V.-L. Nguyen, S. Destercke, E. Hüllermeier, Epistemic uncertainty sampling, in: International Conference on Discovery Science (DS), 2019, pp. 72–86.

[43] L. Breiman, Random forests, Machine Learning 45 (1) (2001) 5–32.

[44] B. Lakshminarayanan, A. Pritzel, C. Blundell, Simple and scalable predictive uncertainty estimation using deep ensembles, in: Advances in Neural Information Processing Systems (NeurIPS), Vol. 30, 2017, pp. 6402–6413.

[45] C. Chen, A. Liaw, L. Breiman, Using random forest to learn imbalanced data, Tech. Rep. 666, University of California, Berkeley (2004).

[46] X.-Y. Liu, J. Wu, Z.-H. Zhou, Exploratory undersampling for class-imbalance learning, IEEE Transactions on Systems, Man, and Cybernetics, Part B 39 (2) (2009) 539–550.

[47] J. Vanschoren, J. Van Rijn, B. Bischl, L. Torgo, OpenML: Networked science in machine learning, ACM SIGKDD Explorations Newsletter 15 (2) (2014) 49–60.

[48] T. Chen, C. Guestrin, XGBoost: A scalable tree boosting system, in: KDD, 2016, pp. 785–794.

[49] F. Wilcoxon, Individual comparisons by ranking methods, Biometrics Bulletin 1 (6) (1945) 80–83.

[50] J. Demšar, Statistical comparisons of classifiers over multiple data sets, Journal of Machine Learning Research 7 (2006) 1–30.

[51] C. Guo, G. Pleiss, Y. Sun, K. Q. Weinberger, On calibration of modern neural networks, in: International Conference on Machine Learning (ICML), 2017, pp. 1321–1330.

[52] D. Dablain, B. Krawczyk, N. V. Chawla, DeepSMOTE: Fusing deep learning and SMOTE for imbalanced data, IEEE Transactions on Neural Networks and Learning Systems 34 (2023) 6390–6404.